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Dynamic response and stability of a flapping foil in a dense and viscous fluid

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It is important to understand and accurately predict the static and dynamic response and stability of flexible hydro/aero lifting bodies to ensure their structural safety, to facilitate the design/optimization of new/existing concepts, and to test the feasibility of using advanced materials. The present study investigates the influence of solid-to-fluid added mass ratio ($\sqrt{\mu_b}$) and viscous effects on the fluid-structure interaction (FSI) response and stability of a flapping foil in incompressible and turbulent flows using a recently presented efficient and stable numerical algorithm in time-domain, which couples an unsteady Reynolds Average Navier-Stokes solver with a two degrees-of freedom structural model. The new numerical coupling method is able to stably and accurately simulate the FSI behavior of light foils in dense fluids: a limit which is known to be numerically difficult to study with classical FSI coupling methods. The studied FSI responses include static/dynamic divergence and flutter instabilities, which are compared with inviscid, linear potential theory predictions obtained with both time and frequency domain formulations, as well as with several published experimental data. In general, the results show that the critical reduced flutter velocities and reduced divergence velocities both decrease as $\sqrt{\mu_b}$ decreases, and are captured with good accuracy using the viscous FSI solver for a wide range of relative mass ratios that are typical to air/hydrofoils. The comparative analyses showed that the classic frequency-domain linear potential theory is severely unconservative for predicting the flutter velocity for cases with $\sqrt{\mu_b} < 3$: this includes the typical operating conditions of most marine and biomedical lifting devices, where the fluid forces are comparable to the solid forces, and strong nonlinear interactions may develop. In addition, the viscous FSI solver is shown to correctly predict the experimentally reported critical divergence speed of a light foil in a dense fluid for a case where the classical potential theory predicts an infinite divergence speed as the foil's elastic axis (E.A.) coincided with the aerodynamic center. The results show that static/dynamic divergence will occur before flutter for light hydrofoils with an E.A. downstream of the center of pressure. However, for high solid-to-fluid added mass ratios ($\sqrt{\mu_b} > 2$), flutter tends to occur prior to divergence. In addition, in between the regions governed by static divergence ($\sqrt{\mu_b} < 1$) and flutter ($\sqrt{\mu_b} > 2$), there is a dynamic divergence region, where the foil deformations oscillate with an increasing mean amplitude, and the oscillation frequency decreases toward zero as the deformation increases; this region could only be captured by using a viscous FSI solver. © *2013 AIP Publishing LLC*. [http://dx.doi.org/10.1063/1.4825136]

I. INTRODUCTION

Aeroelastic/hydroelastic instabilities, such as static/dynamic divergence, and flutter, have been popular research subjects for the past decades. Specifically, static divergence is a linear instability where the fluid induced solid deformations increase with time without oscillations, and is caused

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by the loss of the effective torsional stiffness. Dynamic divergence is similar to static divergence in the sense that the mean deformations increase with time, but is different in that it is a nonlinear, dynamic response with oscillations. For dynamic divergence, the oscillation frequencies decay with increasing deformation. On the other hand, flutter is a dynamic self-excited aeroelastic/hydroelastic instability, for where the solid oscillates with a fixed frequency, and these oscillations grow in time if the instability limit is exceeded; the critical flutter limit is defined for where the effective system damping is zero. Uncontrolled static and/or dynamic instabilities[1–4] can lead to excessive deformation, material failure, accelerated fatigue, vibration, noise issues, and even catastrophic structural failures. On the other hand, intentional flutter can be used to harvest energy from ambient flow.[5,6] The fluid structure interaction (FSI) response and stability are also critical considerations to the function of natural and biomedical devices.[7–9] Hence, it is important to understand and accurately predict the stability conditions of flexible bodies.

Most of the earlier works on FSI problems have been on aerospace or wind engineering structures, where the fluid density is much smaller compared to the solid density, and hence the effects of fluid inertia or damping forces are often ignored. Early works (e.g., Theodorsen,[10] Sears,[11] Garrick[12]) used the linear potential theory to obtain analytical expressions for the aerodynamic lift and moment on two-dimensional thin airfoils undergoing small amplitude harmonic bending and twisting oscillations in uniform inflow. The linear theory (also known as Theodorsen's method) assumed: (1) the total lift force acts at the aerodynamic center, a theoretical location at quarter-chord downstream from the foil leading edge (LE), and (2) the wake behind the foil to consist of shed vortices from the foil trailing edge that move parallel to the inflow and convect downstream at a fixed frequency without any dissipation.

However, the linear potential theory has a few shortcomings. First, as shown in Anderson *et al.*[13] and Munch, Ausoni, and Braun,[14] real wake patterns are typically more complex than the idealized wake mentioned above. The actual wake patterns usually depend on the Strouhal number ($\mathrm{St} = fc/U$) (or reduced frequency ($k = \omega b/U = \pi\,\mathrm{St}$)), Reynolds number ($\mathrm{Re} = Uc/\nu$), and effective angle of attack (α_{eff}), where $\omega = 2\pi f$ is the oscillation frequency, $c = 2b$ is the foil chord length, U is the mean relative inflow speed of the fluid, and ν is the fluid kinematic viscosity. Anderson *et al.*[13] experimentally studied the wake patterns of a 2D pitching and plunging NACA 0012 foil in a water tank for different St and α_{eff}. The foil had a 0.1 m chord and 0.6 m span, and the Re varied from 1.1×10^3 to 4.0×10^4. The water tank size was 30 m × 2.5 m × 1.2 m. For $\mathrm{St} < 0.2$, they observed a wavy wake without distinct vortices with very weak leading edge vortices (LEV) appearing between $0.1 < \mathrm{St} < 0.2$. For $\mathrm{St} > 0.2$, the wake consisted of a von Karman street with LEV. In another study, Munch, Ausoni, and Braun[14] numerically and experimentally studied a pitching hydrofoil (NACA 0009) at different reduced frequencies. The k varied from 0.02 to 100 and the Re varied from 0.5×10^6 to 1.5×10^6 with a 5° maximum angle of attack. For $k \leq 4$ (or $\mathrm{St} \leq 1.27$), the wake pattern consisted two undulating shear layers with opposite strengths; while for $k \geq 12$ (or $\mathrm{St} \geq 3.82$), the wake was observed to be composed of distinct point vortices.

Previous researchers have shown that linear potential flow theory is inadequate to model FSI problems with strong viscous effects. Viscous forces have a greater influence on light and flexible lifting surfaces in water/blood/polymer solutions than on lifting surfaces in air because of the higher density and viscosity. Mandre and Mahadevan[15] presented a unified theory of flutter corresponding to the cases of viscous and inviscid flows. Ducoin and Young[16] calculated the static divergence limit of a spanwise flexible hydrofoil (NACA 66) while considering viscous effects, and showed that viscous effects help to delay divergence. Furthermore, Poirel, Harris, and Benaissa,[17] Poirel and Yuan,[18] and Ducoin and Young[16] have shown that pressure fluctuations generated by the laminar separation bubble and/or unsteady vortex shedding can lead to hydrofoil vibrations, which can cause unwanted noise and vibration issues, as well as potential catastrophic failure of the structure if flutter develops. Apart from foil geometries, Connell and Yue[19] and Akcabay and Young[6] numerically examined the dynamic response and stability of cantilever beams in viscous and axial flows. Akcabay and Young[6] validated their numerical predictions with various experimental data. They found that viscous effects affect the beam tension, and both effects tend to delay flutter, reduce the vibration amplitudes, and change the oscillation modes. They also found that viscous effects are more important for light beams in heavy fluids due to the higher importance of the fluid forces relative to the solid forces.

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Flutter foil experiments in air have been conducted by many, but only few have investigated relative solid-to-fluid density and viscous effects. Woolston and Castile[20] measured the flutter velocities and frequencies of a NACA 16-010 foil with a 0.305 m chord length and a 1.219 m span length, which was made of Balsa wood in a wind tunnel where the fluid density was varied by using different mixtures of Freon 21 and air. The wind tunnel had a 0.61 m $\times$ 1.37 m cross section; the Re ranged from 0.27 $\times$ 10^6 to 3.00 $\times$ 10^6; and the solid-to-fluid added mass ratio, $\sqrt{\mu_b}$, ranged from 1.20 to 13.81. They found that the frequency-domain linear theory predictions underestimated the flutter boundaries (i.e., predicted lower critical flutter velocities, which were overly conservative) for high solid-to-fluid added mass ratios and overestimated the flutter boundaries (unconservative) for low solid-to-fluid added mass ratios. Thomas, Dowell, and Hall[21] numerically studied viscous effects with a time-domain solution technique, and compared their predictions with their own experimental measurements from a NLR 7301 airfoil with a 0.3 m chord length and a 1.0 m span length, that is undergoing pitching and plunging motions at Re = 1.73 $\times$ 10^6 and $\sqrt{\mu_b}$ = 13.11 in a 1 m $\times$ 1 m cross section wind tunnel. While their numerical results predicted lower reduced flutter velocities and higher reduced flutter frequencies compared to the experiments, they reported that their time-domain viscous solution was closer to the experimental results than the more commonly used inviscid, frequency-domain solutions.

In contrast to aeroelastic instability studies, very few studies have focused on hydroelastic instabilities, where the influence of fluid inertia and damping, which are difficult to quantify, are no longer negligible. Flutter experiments of hydrofoils in water are difficult and expensive to conduct; this is due to the relatively high fluid loads, the challenge with finding suitable model-scale materials, the typically lower maximum attainable speeds in water tunnels or towing tanks compared to wind tunnels, and potential complications due to cavitation and free surface effects while operating in water. Note that towing tank tests can capture free surface effects, but cannot properly scale the cavitation physics. On the other hand, while cavitation tunnels can simulate the proper cavitation phenomenon, they cannot correctly scale the free surface effects unless it is a depressurized free-surface cavitation tunnel or a depressurized towing tank facility.

Abramson and Ransleben[22] experimentally examined the flutter instability of a fully submerged hydrofoil and compared their measurements with theoretical results obtained using both the 2D Theodorsen[10] method and the 3D Reissner and Stevens[23] method; Reissner and Stevens[23] method is also known as an extension to Theodorsen's method that accounts for finite span effects. Abramson and Ransleben's[22] experiment involved a lead NACA 16-012 hydrofoil with a 0.3048 m chord length and 1.524 m span length in a towing tank. The critical Re where flutter occurred was 6.70 $\times$ 10^6 for $\sqrt{\mu_b}$ = 0.99. Chu and Abramson[24] calculated the flutter velocity of a fully submerged subcavitating hydrofoil using: (1) the Reissner and Stevens[23] theory, (2) the Yates[25] theory, and (3) the modified Reissner and Stevens theory and Yates theory with a generalized Kutta condition. The Yates[25] method modified Theodorsen's method by correcting the lift curve slope, aerodynamic center, and three-dimensional effects. As for the generalized Kutta condition, a semi-empirical factor was applied to correct the trailing edge tangential velocity. Chu and Abramson[24] found that results obtained with the generalized Kutta condition yielded good agreement with the experimental flutter velocity of Abramson and Ransleben,[22] and the addition of structural damping helped to improve the accuracy of the predicted flutter velocity. Abramson[1] reviewed the literature on hydrofoil flutter with a special focus on the effects of geometric parameters, sweep angle, cavitation, and the submerged depth beneath the free-surface. They reported that the theoretical predictions were very sensitive to the aero/hydrodynamic formulations, geometric parameters, and solid-to-fluid added mass ratio.

Cieslowski and Besch[26] performed experiments at the Naval Ship Research and Development Center (NSRDC) for non-cavitating flows in both a high-speed towing basin and a variable-pressure water tunnel. The towing basin was 904.65 m long, 6.40 m wide, and 3.05 m deep, and had a maximum attainable speed of 30.87 m/s. The water tunnel was a closed circuit system with open and close jet test sections with a maximum attainable speed of 25.72 m/s. The model was designed to investigate the bending and twisting motions of a titanium NACA 16-209 hydrofoil with a 0.18 m chord and a 0.53 m span. They changed the relative mass and moment of inertia ratios by varying the eccentric weights and the radius of gyration; the Re ranged from 2.06 $\times$ 10^6 to 3.11 $\times$ 10^6,

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and the solid-to-fluid added mass ratio ($\sqrt{\mu_b}$) ranged from 1.33 to 3.31. Cieslowski and Besch[26] found that both the solid-to-fluid added mass ratio and foil submergence depth (in the high-speed towing basin) significantly affected the critical flutter velocities. They noted that theoretical results using frequency-domain technique were unconservative (over-predicted the flutter speeds) at low solid-to-fluid added mass ratios. In the towing-tank experiments, the flutter results were observed to approach the theoretical values with an increased submergence depth.

Besch and Liu[27] assessed the flutter and divergence velocities of hydrofoils in water with different structural densities. They used a few different NACA 16-012 hydrofoils made of lead and tin, epoxy and lead powder, epoxy and glass micro balloons. These experiments were performed in a water tunnel, which had a 25.72 m/s maximum speed. In their study, the models with solid-to-fluid added mass ratios of 0.67, 0.63, and 0.45 failed due to large bending deformations before flutter was encountered. They found that the Yates[25] theory over-predicted the flutter velocity for low solid-to-fluid added mass ratios, and hence is unconservative.

As illustrated in the literature review above, the solid-to-fluid added mass ratio has a significant impact on the foil stability boundary, but very limited experimental studies exist in the low solid-to-fluid added mass ratio regime and accurate prediction of the hydroelastic response and stability in this region is still lacking. Therefore, the objective of this work is to investigate the influence of the solid-to-fluid added mass ratio on the FSI response and stability of a flapping foil. For simplicity, the influence of cavitation and free surface are not considered in this work, but will be addressed in future studies.

II. NUMERICAL MODEL

This work numerically investigates the fluid-induced dynamics and stability of a two-dimensional, flexible, NACA 16-010 foil with both a viscous and an inviscid fluid flow model coupled with a two degrees-of-freedom (2-DOF) solid model with a time-domain solution. The predictions are compared with various reported experimental and theoretical results in the literature.

A. Solid model

The flexible foil as shown in Fig. 1 is modeled as a chordwise rigid, 2-DOF system with bending and twisting flexibilities. In Fig. 1, the bending deformation, $h(t)$, is defined positive upwards, and the twisting deformation, $\theta(t)$, is defined positive counterclockwise, and both are defined about the elastic axis (E.A.). The E.A. is located at a distance ab toward the foil trailing edge from the mid-chord (a represents the fraction of the semi-chord b). The chord length is $c = 2b$ and the center of gravity (C.G.) is located at a distance $x_\theta b$ from the E.A. toward the foil trailing edge. A.C. denotes the theoretical aerodynamic center, and ec is the distance between the E.A. and the A.C. which is positive towards the foil leading edge.

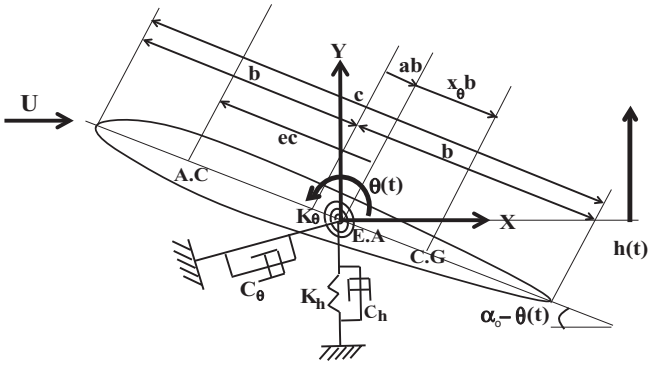


FIG. 1. A two degrees-of-freedom model describing the bending $h(t)$ and the twisting $\theta(t)$ degrees of freedom of a chordwise rigid foil.

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The 2-DOF solid equations of motion can be written as in Eq. (1):

$$\begin{bmatrix} m & S_\theta \\ S_\theta & I_\theta \end{bmatrix} \begin{Bmatrix} \ddot{h} \\ \ddot{\theta} \end{Bmatrix} + \begin{bmatrix} C_h & 0 \\ 0 & C_\theta \end{bmatrix} \begin{Bmatrix} \dot{h} \\ \dot{\theta} \end{Bmatrix} + \begin{bmatrix} K_h & 0 \\ 0 & K_\theta \end{bmatrix} \begin{Bmatrix} h \\ \theta \end{Bmatrix} = \begin{Bmatrix} L_{\text{fluid}} \\ M_{\text{fluid}} \end{Bmatrix}, \tag{1}$$

where m, $S_\theta(= mx_\theta b)$, and $I_\theta(= mr_\theta^2 b^2)$ are, respectively, the structural mass, the static imbalance, and the structural moment of inertia defined about the E.A., while r_θ is the nondimensional radius of gyration. If $\omega_h(= 2\pi f_h)$ and $\omega_\theta(= 2\pi f_\theta)$ are the natural bending and twisting frequencies in vacuum, then $C_h(= 2m\omega_h\zeta_h)$ and $C_\theta(= 2I_\theta\omega_\theta\zeta_\theta)$ are the structural damping values for the bending and twisting motions, respectively; ζ_h and ζ_θ are the nondimensional structural bending and twisting damping coefficients. Similarly, $K_h(= m\omega_h^2)$ and $K_\theta(= I_\theta\omega_\theta^2)$ are the structural bending and torsional stiffness values. The overhead dots indicate Lagrangian time derivatives and hence h, $\dot{h}$, and $\ddot{h}$ are, respectively, the bending displacement, velocity, and acceleration, while θ, $\dot{\theta}$, and $\ddot{\theta}$ are, respectively, the twisting displacement, velocity, and acceleration. L_{fluid} and M_{fluid} are the fluid lift and moment acting on the foil. Note that M_{fluid} is defined positive counterclockwise about the E.A. Equation (1) can be rewritten in a condensed notation as

$$\mathbf{M}_s\ddot{\mathbf{X}} + \mathbf{C}_s\dot{\mathbf{X}} + \mathbf{K}_s\mathbf{X} = \mathbf{F}_{\text{fluid}}, \tag{2}$$

where $\mathbf{M}_s$, $\mathbf{C}_s$, and $\mathbf{K}_s$ are, respectively, the structural inertial, damping, and stiffness matrices, while $\mathbf{X}$, $\dot{\mathbf{X}}$, and $\ddot{\mathbf{X}}$ are, respectively, the structural displacement, velocity, and acceleration vectors with $\mathbf{X} = [h, \theta]^T$. Furthermore, $\mathbf{F}_{\text{fluid}} = [L_{\text{fluid}}, M_{\text{fluid}}]^T$ is the fluid force vector. The current simulations solve Eq. (2) by using a second-order accurate Crank-Nicholson method presented in Johansen.[28]

B. Inviscid FSI model

1. Inviscid fluid model

The inviscid fluid model uses the potential flow solution (assuming an inviscid, irrotational, and incompressible fluid), through Theodorsen's approximation given in Theodorsen,[10] Sears,[11] Garrick.[12] Theodorsen's approach is derived for a thin foil undergoing small harmonic pitching and plunging motions. As the foil section oscillates, it generates a horizontal vortex sheet at its trailing edge without any thickness, which reduces the circulation and lift force on the foil through downwash effect. The linear theory finds the resultant lift force to act at the aerodynamic center (A.C.), which is at quarter-chord downstream from the foil LE.

The inviscid fluid force is decomposed into static ($\mathbf{F}^T_{\text{static}}$) and dynamic ($\mathbf{F}^T_{\text{fluid}}$) parts as shown in Eq. (3):

$$\mathbf{F}_{\text{fluid}} = \mathbf{F}^T_{\text{static}} + \mathbf{F}^T_{\text{fluid}}. \tag{3}$$

The static fluid force, $\mathbf{F}^T_{\text{static}}$, is the fluid force per unit span acting on the foil due to the nonsymmetrical foil geometry and the initial alignment of the foil with respect to the incoming fluid flow (at this position, the bending and torsional elastic forces and hence h and θ are assumed to be zero); it is as given in Eq. (4):

$$\mathbf{F}^T_{\text{static}} = \begin{Bmatrix} L_{\text{static}} \\ M_{\text{static}} \end{Bmatrix} = \begin{Bmatrix} \frac{dC_L}{d\alpha} qc(\alpha_o - \alpha_{Lo}) \\ qc^2 C_{M_{AC}} - L_{\text{static}} ec \end{Bmatrix}, \tag{4}$$

where $dC_L/d\alpha = 2\pi$ is the theoretical slope of the nondimensional lift vs. angle of attack curve, $q = \frac{1}{2}\rho_f U^2$ is the dynamic fluid pressure with U being the relative inflow velocity and ρ_f as the fluid density, while α_o and α_{Lo} are, respectively, the initial angle of attack (which are positive clockwise) and the angle of attack at which the lift force is zero. $C_{M_{AC}}$ is the moment coefficient about the A.C. $e = \frac{1}{4} + \frac{a}{2}$ is the nondimensional distance (as a fraction of c) from the E.A. to the A.C., which is positive toward the foil LE. Note that for a symmetric foil with no camber (such as the current NACA 16-010 foil), α_{Lo} and $C_{M_{AC}}$ are both zero.

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Theodorsen[10] expresses the inviscid fluid lift and moment on the foil as

$$\mathbf{F}^T_{\text{fluid}} = \begin{Bmatrix} L^T_{\text{fluid}} \\ M^T_{\text{fluid}} \end{Bmatrix} = -[\mathbf{M}^T_{\text{f}}]\ddot{\mathbf{X}} - [\mathbf{C}^T_{\text{f}}]\dot{\mathbf{X}} - [\mathbf{K}^T_{\text{f}}]\mathbf{X}, \tag{5}$$

where $\mathbf{M}^T_{\text{f}}$, $\mathbf{C}^T_{\text{f}}$, and $\mathbf{K}^T_{\text{f}}$ are, respectively, Theodorsen's fluid inertial, damping, and stiffness terms, which could be written as

$$\mathbf{M}^T_{\text{f}} = \begin{bmatrix} M^T_{f,hh} & M^T_{f,h\theta} \\ M^T_{f,\theta h} & M^T_{f,\theta\theta} \end{bmatrix} = \begin{bmatrix} \pi\rho_f b^2 & -\pi\rho_f b^3 a \\ -\pi\rho_f b^3 a & \pi\rho_f b^4(\frac{1}{8} + a^2) \end{bmatrix}, \tag{6}$$

$$\mathbf{C}^T_{\text{f}} = \begin{bmatrix} C^T_{f,hh} & C^T_{f,h\theta} \\ C^T_{f,\theta h} & C^T_{f,\theta\theta} \end{bmatrix} = -\pi\rho_f U b^2 \begin{bmatrix} -\frac{2}{b}C(k) & -(1 + (1-2a)C(k)) \\ (2a+1)C(k) & b(\frac{1}{2} - a)(-1 + (2a+1)C(k)) \end{bmatrix}, \tag{7}$$

$$\mathbf{K}^T_{\text{f}} = \begin{bmatrix} K^T_{f,hh} & K^T_{f,h\theta} \\ K^T_{f,\theta h} & K^T_{f,\theta\theta} \end{bmatrix} = \begin{bmatrix} 0 & 2\pi\rho_f U^2 b C(k) \\ 0 & -\pi\rho_f U^2 b^2 (2a+1)C(k) \end{bmatrix}, \tag{8}$$

where $k = \omega b/U$ is the reduced frequency and ω is the system oscillation frequency. The Theodorsen's circulation function, C(k), in Eqs. (7) and (8), is given in Eq. (9); it can be considered as a measure to quantify the wake induced loss of the lift,

$$C(k) = F(k) + iG(k) = \frac{H^2_1(k)}{H^2_1(k) + iH^2_0(k)}, \tag{9}$$

where $H^2_1(k)$ and $H^2_0(k)$ are Hänkel functions (i.e., Bessel function of the third kind).

2. Inviscid fully coupled (FC) FSI model

The inviscid Fully Coupled (FC) FSI model discretizes the equations of motion (EOM), Eqs. (2)–(5) in the time-domain as

$$[\mathbf{M}_{\text{s}} + \mathbf{M}^T_{\text{f}}]\ddot{\mathbf{X}}_{n+1} + [\mathbf{C}_{\text{s}} + \mathbf{C}^T_{\text{f}}]\dot{\mathbf{X}}_{n+1} + [\mathbf{K}_{\text{s}} + \mathbf{K}^T_{\text{f}}]\mathbf{X}_{n+1} = \left\{\mathbf{F}^T_{\text{static}}\right\}_{n+1}, \tag{10}$$

where the subscript n denotes the discrete time-level. Note that the solid and fluid inertial, damping, and restoring force components are all treated implicitly with the inviscid FC FSI formulation; such an implicit treatment is easy to do here as an analytical expression for $\mathbf{F}_{\text{fluid}}$ is available for a foil with bending and twisting degrees of freedom only, and the EOM is linear. The oscillation frequency k, which is needed to calculate the terms in Eqs. (7) and (8) is predicted via an iterative process.

C. Viscous FSI model

1. Viscous fluid model

For the viscous fluid model, the current simulation solves the unsteady Reynolds Averaged Navier-Stokes (uRANS) equations with the $k-\omega$ Shear Stress Transport (SST) turbulence closure model for an incompressible Newtonian fluid with negligible body forces:

$$\rho_f\left(\frac{\partial \mathbf{u}}{\partial t} + \mathbf{u} \circ \nabla\mathbf{u}\right) = -\nabla p + \nabla \circ \left(\mu_{\text{eff}}\left(\nabla\mathbf{u} + \nabla\mathbf{u}^T\right)\right), \tag{11}$$

$$\nabla \circ \mathbf{u} = 0, \tag{12}$$

where $\mathbf{u}$ is the local fluid velocity, t is time, p is the pressure, and μ_{eff} is the effective dynamic viscosity, which depends on the particular turbulence closure model. The simulation solves Eqs. (11) and (12) via the ANSYS-CFX[29] commercial software.

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2. Hybrid coupled (HC) viscous FSI method

The viscous simulations use the Hybrid Coupled (HC) method introduced in Young, Chae, and Akcabay[30] to couple the partitioned viscous computational fluid dynamics (CFD) and computational structure dynamics (CSD) solvers to model flexible structures in dense, incompressible flow. A partitioned FSI coupling method is used for this, because of the conveniences associated with using modular CFD and CSD solvers. Classically, partitioned FSI methods couple the two different CFD and CSD solvers via either the loosely coupled (LC) or tightly coupled (TC) techniques. The LC technique uses the CFD solution at the previous time-step as an input to the CSD solution for the new time-step, while the TC technique iterates between the solutions of the CFD and CSD solvers per each new time-step. Previous works of Belanger, Paidoussis, and de Langre;[33] Mok and Wall;[34] Causin, Gerbeau, and Nobile;[35] and Forster, Wall, and Ramm[36] have shown that traditional methods are not suitable for problems involving light solids in incompressible, heavy fluids, due to numerical instability issues associated with artificial added mass effects, which are caused by time-lag in the exchange of surface tractions and deformations between the fluid and solid solvers.

The discretized EOM of the viscous HC method in time-domain is given in Eq. (13):

$$[\mathbf{M}_\mathrm{s} + \mathbf{M}_\mathrm{f}^T]\ddot{\mathbf{X}}_{n+1}^{i+1} + [\mathbf{C}_\mathrm{s} + \mathbf{C}_\mathrm{f}^T]\dot{\mathbf{X}}_{n+1}^{i+1} + [\mathbf{K}_\mathrm{s} + \mathbf{K}_\mathrm{f}^T]\mathbf{X}_{n+1}^{i+1} = \{\mathbf{F}_\mathrm{CFD}\}_{n+1}^{i} - \left\{\mathbf{F}_\mathrm{fluid}^T\right\}_{n+1}^{i}, \quad (13)$$

where i is the sub-iteration number within each time-step level n. $\mathbf{F}_\mathrm{CFD}$ is the viscous fluid force on the foil that is computed by the CFD solver, which includes both static and dynamic force components. The HC method subtracts the potential flow estimate of the hydroelastic force ($\mathbf{F}_\mathrm{fluid}^T$) from both the left and right sides of the EOM. For a thin 2-D foil, the HC method estimates $\mathbf{F}_\mathrm{fluid}^T$ using Theodorsen's derivation, as given in Eqs. (5)–(8). It should be noted that for more general cases (e.g., complex 3D solids), the potential solution could be obtained from 3D boundary integral methods such as those presented in Young.[37,38]

Equation (13) is solved using a semi-implicit Crank-Nicholson method, which is second-order accurate in time. At the first sub-iteration ($i = 1$) at each time-step, the hydroelastic forces are computed using the displacements from the last iteration ($i = imax$) of the previous time-step, i.e.,

$$\mathbf{X}_{n+1}^{i=1} = \mathbf{X}_{n}^{i=imax}. \quad (14)$$

The HC algorithm flow chart for the HC algorithm is given in Fig. 2. $\mathbf{X}_o = [h_o, \theta_o]^T$ are the initial prescribed displacements. ξ_r is a relaxation factor, γ is a regulator such that $0 < \gamma \ll 1$, ϵ is the error tolerance value to determine convergence within the sub-iterations, and *ITMAX* is the maximum allowed number of sub-iterations. $\mathbf{X}_o$, $\dot{\mathbf{X}}_o$ are, respectively, the initial deformations and speed of deformations. In this study, the commercial ANSYS-CFX[29] software is used as the CFD solver, but other commercial or in-house CFD solvers could be used as well with the HC algorithm. The current simulations use $\xi_r = 1$, $ITMAX = 1$, and will be referred as the viscous Loose Hybrid Coupled (LHC) method from here on.

It should be noted that in a similar approach to the HC method, Mok and Wall;[34] Causin, Gerbeau, and Nobile;[35] Forster, Wall, and Ramm;[36] and Xiao and Batra[39] added the fluid added mass to the structural mass on the left-hand side of the EOM. Similarly, Chen and Wambsganss,[31] Paidoussis,[32] and Young,[37,38] added both the fluid added mass and viscous damping terms to the left-hand side of the EOM. The HC method is also similar to the coupling methods presented by Belanger, Paidoussis, and de Langre;[33] Connell and Yue;[19] and Baek and Karniadakis,[40] which used an estimated $\mathbf{F}_\mathrm{fluid}^T$ to implicitly account for the fluid inertial and/or damping forces. However, the HC method differs from the above methods in that it subtracts the potential flow estimate of all three components (inertial, damping, and stiffness) of the FSI forces from both sides of the EOM, while Belanger, Paidoussis, and de Langre;[33] Connell and Yue;[19] and Baek and Karniadakis[40] only subtracts the added mass and/or added damping term. As shown in Young, Chae, and Akcabay,[30] the HC method is able to significantly deter numerical instability issues due to virtual added mass effects, and to significantly accelerate convergence compared to standard loosely coupled and tightly coupled approaches for modeling of FSI problems involving dense, incompressible, and viscous flow. Additional numerical and experimental validation studies of the LHC method for predicting the

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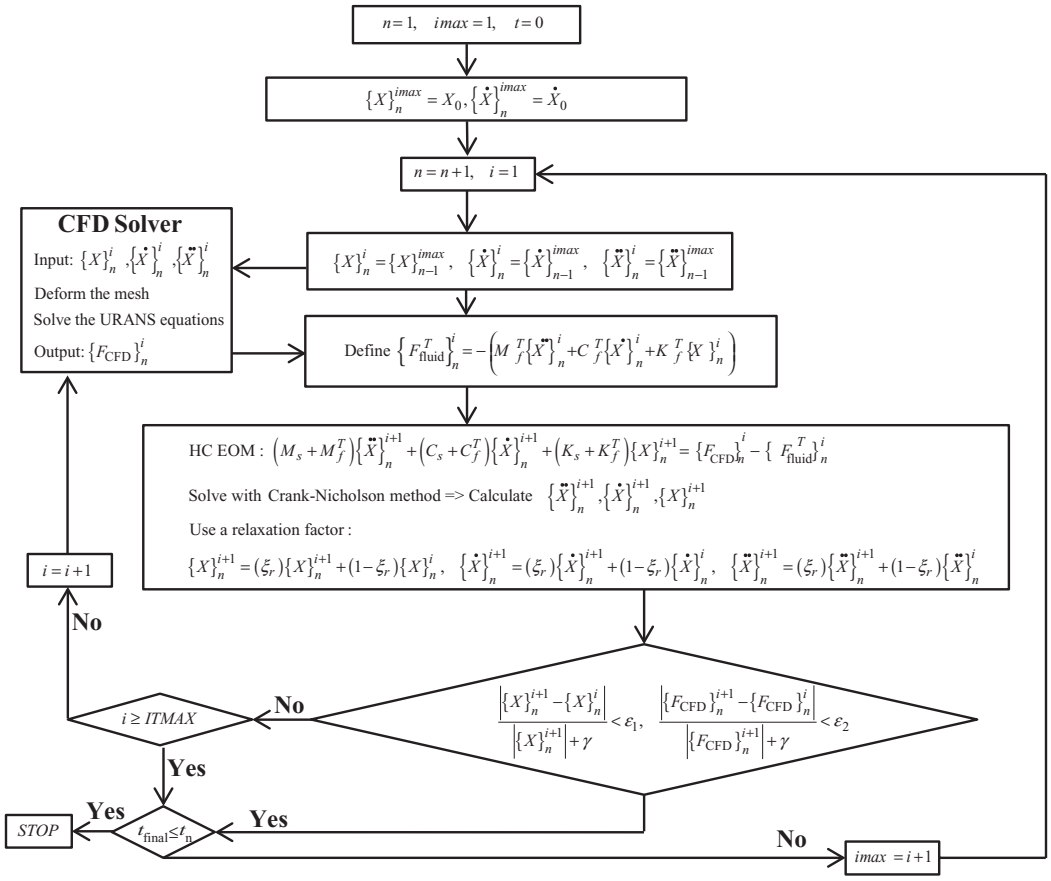


FIG. 2. HC flow chart.

steady-state and dynamic response of flexible hydrofoils in single and multi-phase flows can be found in Akcabay *et al.*[41]

3. Boundary conditions at the fluid-solid interface

The boundary conditions at the fluid-solid interface should satisfy both a kinematic and a dynamic compatibility condition. For the kinematic compatibility condition, the velocity must be locally the same for the fluid and the solid at the common interface. For the dynamic equilibrium condition, the interface forces acting on the fluid and the solid should be equal in magnitude but opposite in direction. For a Newtonian fluid, the total stress tensor ($\boldsymbol{\sigma}$) is

$$\boldsymbol{\sigma} = -p\mathbf{I} + \mu_{\text{eff}}(\nabla\mathbf{u} + \nabla\mathbf{u}^T), \tag{15}$$

where $\mathbf{I}$ is the identity matrix. Therefore, the resulting viscous fluid lift and moment acting on the hydrofoil, $\mathbf{F}_{\text{CFD}}$ in Eq. (13), are computed as follows:

$$\mathbf{F}_{\text{CFD}} = \begin{Bmatrix} L_{\text{fluid}} \\ M_{\text{fluid}} \end{Bmatrix} = \begin{Bmatrix} \oint_A (\mathbf{j} \circ \boldsymbol{\sigma}\mathbf{n}) \mathrm{d}A \\ \oint_A (\mathbf{r} \times \boldsymbol{\sigma}\mathbf{n}) \mathrm{d}A \end{Bmatrix}, \tag{16}$$

where $\mathbf{n}$ is a unit normal vector on the foil surface pointing out of the solid and into the fluid, $\mathbf{j}$ is a unit vector along the Y direction as shown in Fig. 1, and $\mathbf{r}$ is a vector from the solid elastic-axis to a point on the foil surface. The integrations are taken along the closed foil surface, A.

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TABLE I. Physical parameters.

Parameter	Unit	Current simulation	Woolston and Castile	Besch and Liu
Foil	(-)	NACA 16-010	NACA 16-010	NACA 16-012
Material	(-)	Different Balsa woods	Balsa wood	Lead and tin, epoxy and lead powder, Epoxy and glass microballoon
Fluid	(-)	Water, mixtures of Freon 21 and air	Mixtures of Freon 21 and air	Water
c	(m)	0.305	0.305	0.152
s	(m)	2D sim.	1.219	0.381
Maximum thickness	(-)	$0.1c$	$0.1c$	$0.12c$
α_o	($^\circ$)	0	0	0
a	(-)	-0.218 and -0.5	-0.218	-0.5
x_θ	(-)	0.068 and 0.524	0.068	0.523–0.524
r_θ	(-)	0.403 and 0.709	0.403	0.709–0.713
E	(GPa)	1.80 and 2.34	1.80	1.59–1.63
ν	(-)	0.36 and 0.431	0.36	N/A
m	(kg/m)	0.63–286.06	0.63	3.94–17.55
ρ_f	(kg/m^3)	0.5–1000	0.08–3.18	1000
ν_f	(m^2/s)	1.0×10^{-6}–1.12×10^{-5}	2.85×10^{-6}–1.19×10^{-4}	1.0×10^{-6}
Re	(-)	0.35×10^6–4.27×10^6	0.41×10^6–1.84×10^6	1.94×10^6–2.82×10^6
f_h	(Hz)	0.72–17.65	15.29	11.09–22.49
f_θ	(Hz)	1.74–52.11	36.92	35.09–68.75

III. DYNAMIC FSI RESPONSE AND STABILITY OF A FLAPPING FOIL IN LIGHT AND DENSE FLUIDS

A. Physical and numerical model setup

The physical parameters used in the simulations are listed in Table I. Note that since the current simulations are conducted with a 2D fluid solver, so the mass, stiffness, and damping values of the foils in Table I are given per unit span length. In all the simulations, the structural damping is ignored ($C_h = 0$ and $C_\theta = 0$). The numerical mesh for the CFD solver is shown in Fig. 3. The numerical mesh consists of unstructured triangular elements everywhere except at the boundary layer region near the foil, which is discretized with a structured mesh. The smallest element near the foil surface satisfies $y^+ \approx 1$. The mesh is refined near the foil leading and trailing edges, as well as in the wake region, in order to capture the flow details. The boundary conditions are uniform flow with 2.95% turbulence intensity at the inlet (left-end of the numerical domain shown in Fig. 3), symmetry conditions at the top and bottom of the fluid domain, solid-wall boundary condition on the foil surface, and prescribed pressure boundary condition at the outlet. The mesh elements are deformed according to the computed bending and twisting displacements of the hydrofoil at each time-step. The numerical simulations are initialized from the steady-state solution of the fluid flow around the foil with a $\theta_o = -2^\circ$ initial twist angle and $h_o = 0$.

B. Convergence studies

Spatial and temporal convergence studies were conducted for the case corresponding to a Balsa wood foil in a Freon 21 - air mixture with $\mathrm{Re} = Uc/\nu = 1.24 \times 10^6$, $\sqrt{\mu_b} = 2.79$, and $U^* = 1.3$; the relevant experimental description is available at Woolston and Castile.[20] Note that the solid-to-fluid

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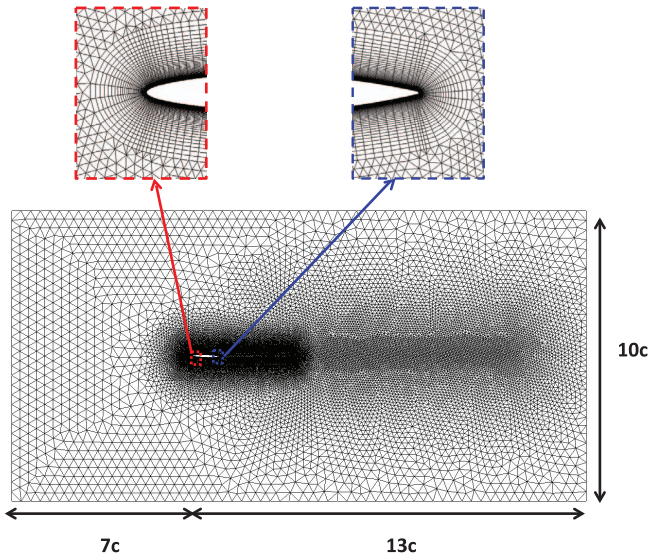


FIG. 3. The 2D mesh, which is discretized with approximately 76 000 elements and 48 000 nodes.

added mass ratio ($\sqrt{\mu_b}$) and reduced velocity (U^*) are defined as

$$\sqrt{\mu_b} = \sqrt{\frac{m}{\pi \rho_f b^2}}, \tag{17}$$

$$U^* = \frac{U}{\omega_\theta b}, \tag{18}$$

where ω_θ is the natural twisting frequency of the foil in vacuum. For the convergence study, the natural bending and twisting frequencies of this foil in vacuum are, respectively, $f_h = 15.29$ Hz and $f_\theta = 36.92$ Hz, while the corresponding natural bending and twisting frequencies in fluid, $f_{h\,-\,fluid}$ and $f_{\theta\,-\,fluid}$, are, respectively, 14.39 Hz and 26.31 Hz. These wetted natural frequencies are estimated by using Eqs. (19) and (20):

$$\omega_{h-fluid} = 2\pi f_{h-fluid} = \sqrt{\frac{K_h + K^T_{f,hh}}{m + M^T_{f,hh}}}, \tag{19}$$

$$\omega_{\theta-fluid} = 2\pi f_{\theta-fluid} = \sqrt{\frac{K_\theta + K^T_{f,\theta\theta}}{I_\theta + M^T_{f,\theta\theta}}}, \tag{20}$$

where $K^T_{f,hh}$, $K^T_{f,\theta\theta}$, $M^T_{f,hh}$, and $M^T_{f,\theta\theta}$ are predicted from Eqs. (6) and (8).

In all the convergence results shown in this subsection, the time-history plots are normalized by the natural twisting period of the foil in fluid, $T = 2\pi/\omega_{\theta\,-\,fluid} = 0.04$s.

For spatial convergence study, three different meshes are used with $\Delta t/T \approx 0.01$ ($\Delta t = 4 \times 10^{-4}$ s). The coarse mesh approximately contains 34 000 nodes and 52 000 elements; the medium mesh approximately contains 48 000 nodes and 76 000 elements; and the fine mesh approximately contains 107 000 nodes and 184 000 elements. Figure 4 shows that the predicted time-histories of the bending and twisting deformations with the medium mesh are sufficiently close with the predictions evaluated with the fine mesh. Hence, from here on, the medium mesh is used to calculate the viscous FSI responses.

For temporal convergence study, the predicted time-histories of the bending and twisting deformations using three different time-step sizes ($\Delta t/T \approx 0.02$, 0.01, and 0.003) on the "medium mesh" (see the previous paragraph) are compared in Fig. 5. Note in Fig. 5 that the predictions obtained

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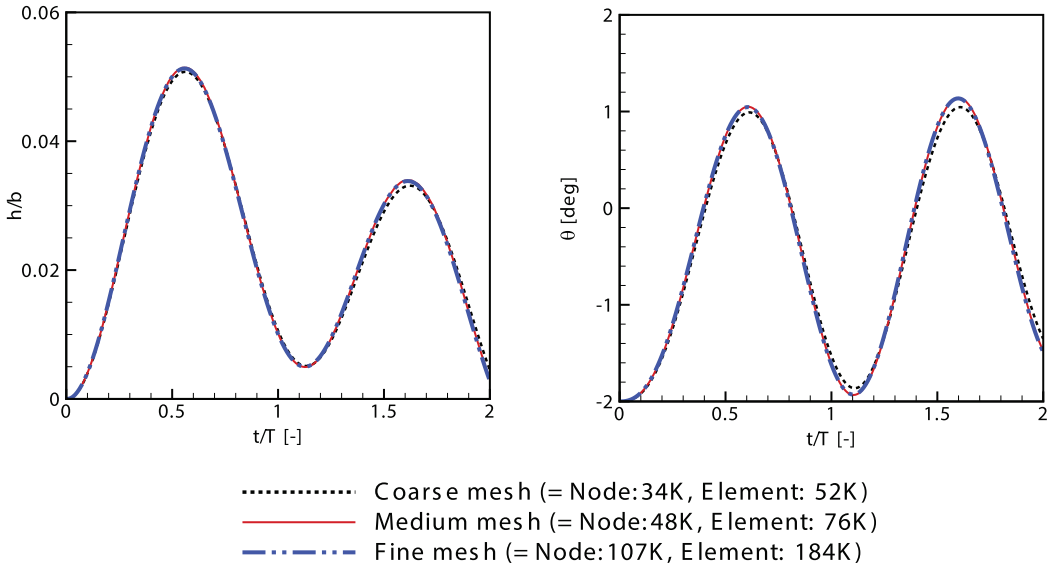


FIG. 4. Spatial convergence studies using three different meshes (coarse mesh: 34 K nodes and 52 K elements, medium mesh: 48 K nodes and 76 K elements, and fine mesh: 107 K nodes and 184 K elements) at Re $= 1.24 \times 10^6$, $\sqrt{\mu_b} = 2.79$, $U^* = 1.3$, and $\Delta t/T \approx 0.01$ with $y^+ \approx 1$.

using the two smallest time-steps are close. Hence, in the simulations shown from here on, a time-step size of $\Delta t = 4 \times 10^{-4}$ s corresponding to $\Delta t/T \leq 0.01$ will be used. This corresponds to a $CFL(= U\Delta t/\Delta x)$ number between 0.07 and 6.70 for the various cases shown in the rest of this paper.

In addition to these convergence studies, the simulations are validated through comparison with various experimental data, which will be shown later in Figs. 7 and 8.

C. Prediction of flutter, static divergence, and dynamic divergence

In all the simulations, the flutter boundary is determined by seeking the minimum critical inflow velocity at which the total damping of the vibrating system, ζ_t, is zero. Note that the total damping

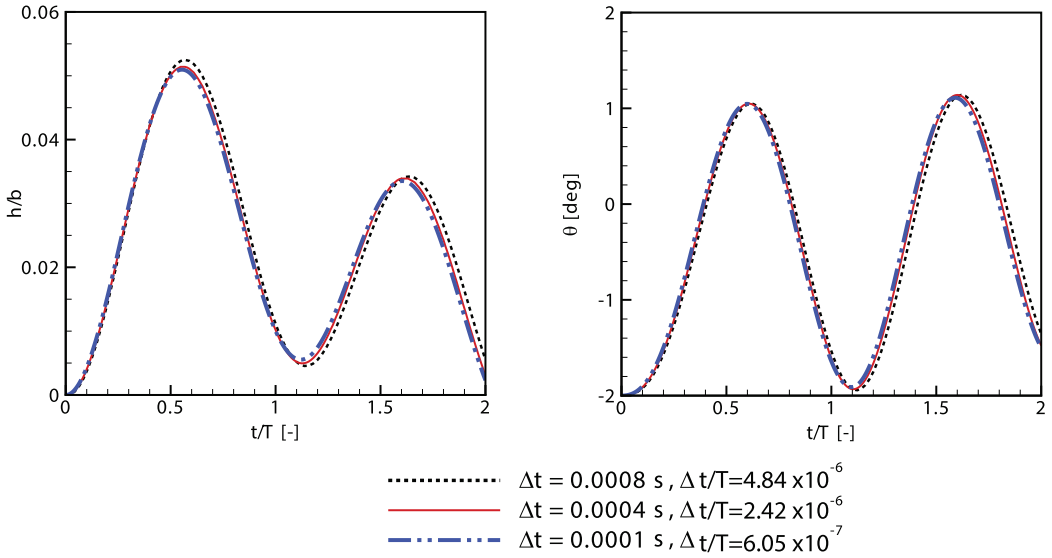


FIG. 5. Temporal convergence studies using three time-steps ($\Delta t/T \approx 0.02$, 0.01, and 0.003) on the "medium mesh" (48 K nodes and 76 K elements) at Re $= 1.24 \times 10^6$, $\sqrt{\mu_b} = 2.79$, $U^* = 1.3$ with $y^+ \approx 1$.

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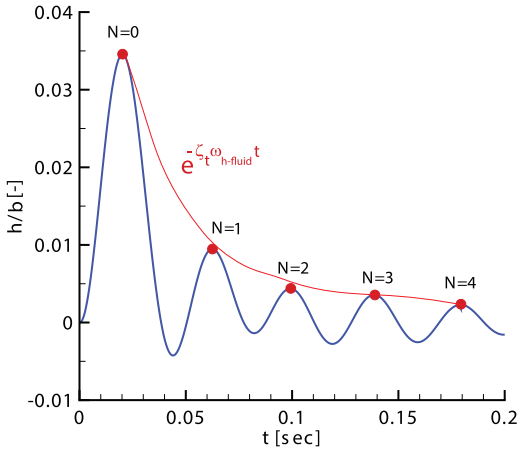


FIG. 6. Sample illustration of determining the total damping coefficient, ζ_t from a given bending deformation time-history result. This result is from the viscous computations with Re $= 5.05 \times 10^6$, $\sqrt{\mu_b} = 4.18$, and $U^* = 1.7$.

is the sum of the structural and fluid damping. The procedure used to determine ζ_t is demonstrated through Fig. 6, and in Eqs. (21) and (22), where N is the number of the peak points between $h(t)$ and $h(t + NT)$, δ is the logarithmic decrement, and T is the system vibration period. Figure 6 is a sample bending deformation plot, which was evaluated using the viscous LHC simulations with Re $= 5.05 \times 10^6$, $\sqrt{\mu_b} = 4.18$, and $U^* = 1.7$. For a given $\sqrt{\mu_b}$ and Re, once ζ_t is determined for a few different U^* values, the flutter boundary is then determined through interpolating the critical U^* for which ζ_t is zero,

$$\delta = \frac{1}{NT} ln \frac{h(t)/b}{h(t + NT)/b}, \tag{21}$$

$$\zeta_t = \frac{\delta}{\sqrt{4\pi^2 + \delta^2}}. \tag{22}$$

The divergence velocity, U_d, is the critical speed at which the effective torsional stiffness (i.e., sum of both structural stiffness and fluid stiffness) of the foil becomes zero. From the balance of fluid-induced static moment and the elastic restoring moment, the linear theory estimates U_d as

$$U_d = \sqrt{\frac{2K_\theta}{\rho_f e c^2 \frac{\partial C_L}{\partial \alpha}}}, \tag{23}$$

where the linear theory takes $e = 1/4 + a/2$ and $\partial C_L/\partial \alpha = 2\pi$. Note have that for a given solid, the dimensional value of U_d decreases with increasing ρ_f according to Eq. (23), but U_d does not vary with ρ_s as long as K_θ remains the same.

In the viscous LHC method solution with a time-domain solution technique, the static divergence velocity is determined as the speed at which the deformations monotonically increase without an obvious oscillation frequency, and the dynamic divergence velocity is determined as the speed for when the mean deformations grow with a decaying oscillation frequency. It should be mentioned here that dynamic divergence is a nonlinear phenomenon,[2–4] and it is only predicted in this work using the LHC viscous FSI solver through a time-domain solution method.

D. Static and dynamic stability boundaries

Figures 7 and 8 compare the current time-domain [TD] solution of the inviscid FC and viscous LHC simulation results with frequency-domain [FD] solution of the linear potential theory predictions and experimental measurements of the critical divergence velocities, flutter velocities and frequencies given in Woolston and Castile[20] and Besch and Liu[27] for cases with different

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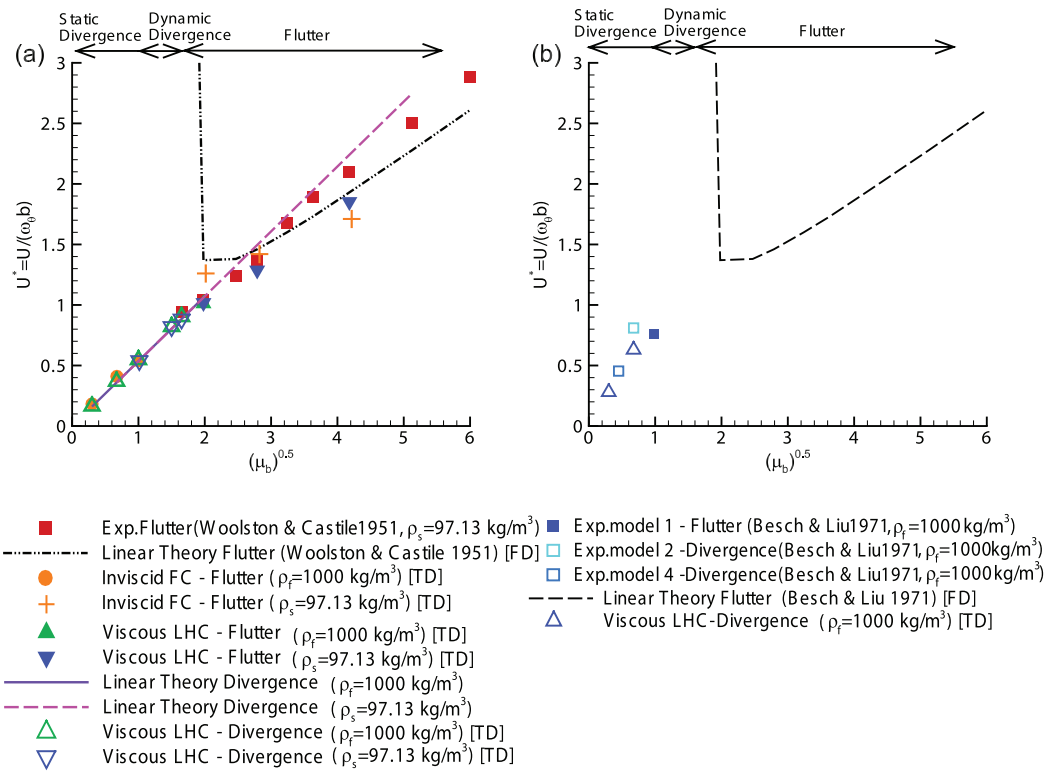


FIG. 7. Static/dynamic divergence and flutter velocity boundaries as a function of the solid-to-fluid added mass ratio ($\sqrt{\mu_b}$) based on (a) Woolston and Castile's[20] data with fixed ρ_f or ρ_s, (b) Besch and Liu's[27] data with varying ρ_s but fixed ρ_f. The linear theory divergence velocities are calculated by Eq. (23); this method predicted $U_d = \infty$ for Besch and Liu's[27] case shown in Fig. 7(b) due to the coincidence between E.A. and A.C., and hence this result is not shown in Fig. 7(b). Note that the closed symbols represent flutter and open symbols represent divergence. The triangle symbols represent cases where $\sqrt{\mu_b}$ is varied with fixed fluid density, and the inverted triangle symbols represent cases where $\sqrt{\mu_b}$ is varied with fixed solid density in the viscous FSI simulations.

solid-to-fluid added mass ratios ($\sqrt{\mu_b}$). The physical parameters used in these experiments are listed in Table I.

Woolston and Castile's[20] experiments cover a $\sqrt{\mu_b}$ range that correspond to the typical operating conditions of light airfoils and Micro Air Vehicles (MAV). The flutter velocities and frequencies in that study were measured for a Balsa wood NACA 16-010 airfoil in Freon 21 and air mixtures with different fluid densities. These experiments were performed in a wind tunnel, which had a circular cross section of 1.37m in diameter. The Re was between 0.27×10^6 and 3.00×10^6. The flutter velocities were determined as the speeds for when the total system damping was projected to be zero.

Besch and Liu's[27] experiments were conducted at typical hydrofoil operating conditions. The relative mass ratios were controlled by using different materials to construct the NACA 16-012 hydrofoil (i.e., lead and tin, epoxy and lead powder, epoxy and glass micro balloons in water). The experiments were performed in a depressurized water tunnel, where the maximum attainable speed was 25.72 m/s. Again, the flutter inception speeds were determined by projecting the speed at which the total damping coefficient was zero.

As shown in Figs. 7(a) and 8, the linear theory and experimental results obtained by Woolston and Castile[20] are in good agreement with each other for $\sqrt{\mu_b} \geq 3$. It is important to note that Woolston and Castile's[20] results are limited to $\sqrt{\mu_b} > 1.5$, due to the use of Balsa wood for the foil, and the limited range of fluid densities that can be achieved by using a Freon 21 and air mixture in a wind tunnel. On the other hand, Besch and Liu's[27] experimental results given in Fig. 7(b) contains data for $\sqrt{\mu_b} \leq 1$ only, since it uses water as the experimental test fluid. It should

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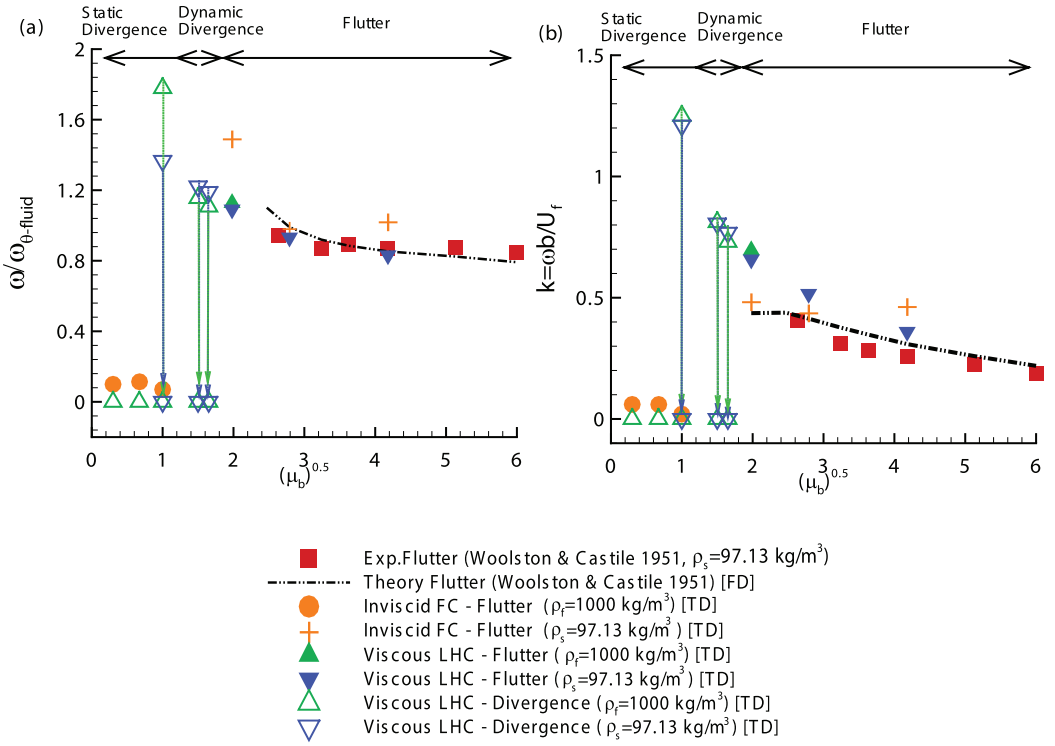


FIG. 8. Static/dynamic divergence and flutter (a) nondimensional frequency ratios and (b) reduced frequencies as a function of solid-to-fluid added mass ratio ($\sqrt{\mu_b}$) based on Woolston and Castile's[20] data with fixed ρ_f or ρ_s. Note that the closed symbols represent flutter and open symbols represent divergence. The triangle symbols represent cases where $\sqrt{\mu_b}$ is varied with fixed fluid density, and the inverted triangle symbols represent cases where $\sqrt{\mu_b}$ is varied with fixed solid density in the viscous FSI simulations.

be noted that Woolston and Castile's[20] linear theory results used the frequency-domain solution technique, while the current inviscid FC and viscous LHC simulations used the time-domain solution technique.

Figure 7 suggests that with decreasing $\sqrt{\mu_b}$, the reduced flutter velocity (U_f^*) and the reduced divergence velocity (U_d^*) will both decrease. According to Eq. (23) for a given foil with a fixed stiffness, the dimensional values of U_d will only decrease as the ρ_f increases (the dimensional U_d is not a function of ρ_s). However, the reduced divergence velocity (U_d^*) decreases with reductions in ρ_s because ω_θ increases with decreases in ρ_s.

The predicted flutter boundary using the current viscous LHC simulations are in good agreement with the experimental results of Woolston and Castile[20] and Besch and Liu.[27] However, as shown in Fig. 7, the frequency-domain linear theory solution predicts higher flutter velocities than the measured values for $\sqrt{\mu_b} < 3$, which is dangerous because that means the actual system will flutter earlier than the theoretical prediction. This discrepancy is because the linear potential theory of flutter assumes small harmonic deformations, which is not necessarily valid for cases with low solid-to-fluid added mass ratios because of the large deformations caused by the high fluid loading, which leads to strong FSI coupling effects. In addition, viscous effects become relatively more important as $\sqrt{\mu_b}$ decreases, where the fluid forces become comparable to the solid forces. Hence, the frequency-domain linear theory is not suitable for flutter predictions of light foils in a dense fluid, such as most hydrofoils. For $1 \leq \sqrt{\mu_b} < 2$ (again in the hydrofoil region), the dynamic divergence and flutter boundaries are very close. The results in Fig. 7 suggest that the viscous LHC method can accurately capture the flutter onset condition for the range of relative mass ratios considered. The time-domain inviscid FC method (in contrast to the linear theory results from Woolston and Castile[20] obtained using a frequency-domain solution method) also yielded good estimates, but a slight over/underestimation of the flutter onset condition is observed because it ignores nonlinear viscous

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FSI effects. However, it is important to point out that the time-domain inviscid FC solver is able to provide a good approximation at a much lower computational cost compared to the viscous LHC method.

Figure 8(a) shows the nondimensional frequency ratios, $\omega/\omega_{\theta-\text{fluid}}$, which is the structural flutter/dynamic-divergence vibration frequency divided by the natural twisting frequency in fluid (Eq. (20)). Figure 8(b) shows the reduced frequency, k, at the flutter boundary. For almost all the cases shown in this work, $k < 1$, and the simulations show that the wake patterns consist of thin, undulating shear layers with opposite signs – as will be shown later in Figs. 15 and 16. Note that the system vibration frequency is zero in the static divergence range ($\sqrt{\mu_b} < 1$), but there are still vibrations at the dynamic divergence range ($1 \le \sqrt{\mu_b} < 2$), where the mean deformations grow in time and the vibration frequency decreases toward zero as the deformation increases. The results show that in the flutter region ($\sqrt{\mu_b} \ge 2$), $\omega/\omega_{\theta-\text{fluid}}$ and k both decrease with increasing $\sqrt{\mu_b}$, and the flutter frequency is slightly less than the wetted natural twisting frequency.

The results in Figs. 7 and 8 suggest that the flutter velocities and frequencies, as well as divergence velocities predicted using the viscous LHC are in good agreement with the measured data.

The trends given in Fig. 7 show that at $\sqrt{\mu_b} = 0$, the flutter velocity is not necessarily zero (as also shown in Caporali and Brunelle[42]), but static divergence will occur earlier than flutter. In addition, for some cases, such as those shown in Fig. 7(b), static divergence may occur at lower speeds than that predicted by linear potential theory because of viscous and nonlinear effects, as well as material failures. Examples of such cases were reported in experimental studies in Besch and Liu[27] and Abramson and Ransleben,[22] and shown on the Fig. 7(b), where the theoretical divergence velocity was infinite because the E.A. coincided with the theoretical A.C. (i.e., $e = 0$ in Eq. (23)); the predicted divergence velocity using the viscous FSI solver was finite, and agreed well with measured values, because viscous effects and flow separation modified the center of pressure (C.P.) from the theoretically assumed location. Nevertheless, it should be noted that the current viscous FSI solver assumes linear elastic material behavior, and does not consider the strain and/or stress limit where material failures may occur. The earlier occurrence of static divergence than flutter at the low solid-to-fluid added mass ratio regime ($\sqrt{\mu_b} < 1$) could be explained as follows: at the low solid-to-fluid added mass ratio regime, the wetted natural frequencies reduce with increased fluid density, which will prolong the structural dynamic response time and increase not only the fluid inertial and damping forces, but also the fluid disturbing force, which acts to lower the effective system stiffness. Note that static divergence is a static failure which occurs when the effective system stiffness is zero, i.e., when the fluid disturbing moment is equal to or higher than the solid restoring moment.

Figure 9 shows the predicted sample time histories for static divergence (Fig. 9(a)), dynamic divergence (Figs. 9(b) and 9(c)), and flutter (Fig. 9(d)) obtained using the viscous LHC method, which are obtained by changing $\sqrt{\mu_b}$ by fixing ρ_f corresponding to the water and varying ρ_s. As shown in Figs. 7 and 8, linear potential theory cannot predict the proper switch-off point between divergence and flutter because there is a dynamic divergence region between flutter and static divergence for $1 \le \sqrt{\mu_b} < 2$, where the mean deformation increases monotonically, and the oscillation frequency decreases as deformation increases, which was captured by the nonlinear viscous FSI solver. Again, it should be emphasized that dynamic divergence is a nonlinear phenomenon, and cannot be predicted with linear methods. Unlike static divergence, dynamic divergence depends on both solid and fluid inertia and damping terms, in addition to the solid restoring force and fluid disturbing force terms. It should be also noted that viscous damping is much higher than inviscid damping for hydrofoils at the low solid-to-fluid added mass ratio regime.

Using the viscous LHC method, the influence of varying ρ_s vs. ρ_f, on the reduced static/dynamic divergence and flutter velocities are shown in Table II. The static divergence (i.e., deformation $\to \infty$ and $k = 0$), dynamic divergence (i.e., mean deformation $\to \infty$ and $k \to 0$), and flutter (i.e., total damping $= 0$ and $k =$ constant) boundaries are distinguished for different solid-to-fluid added mass ratios. The results in Table II show that the reduced velocity (U^*), nondimensional frequency ratio ($\omega/\omega_{\theta\ -\ fluid}$), and reduced frequency (k) for viscous LHC responses are similar for cases with varying ρ_s or ρ_f.

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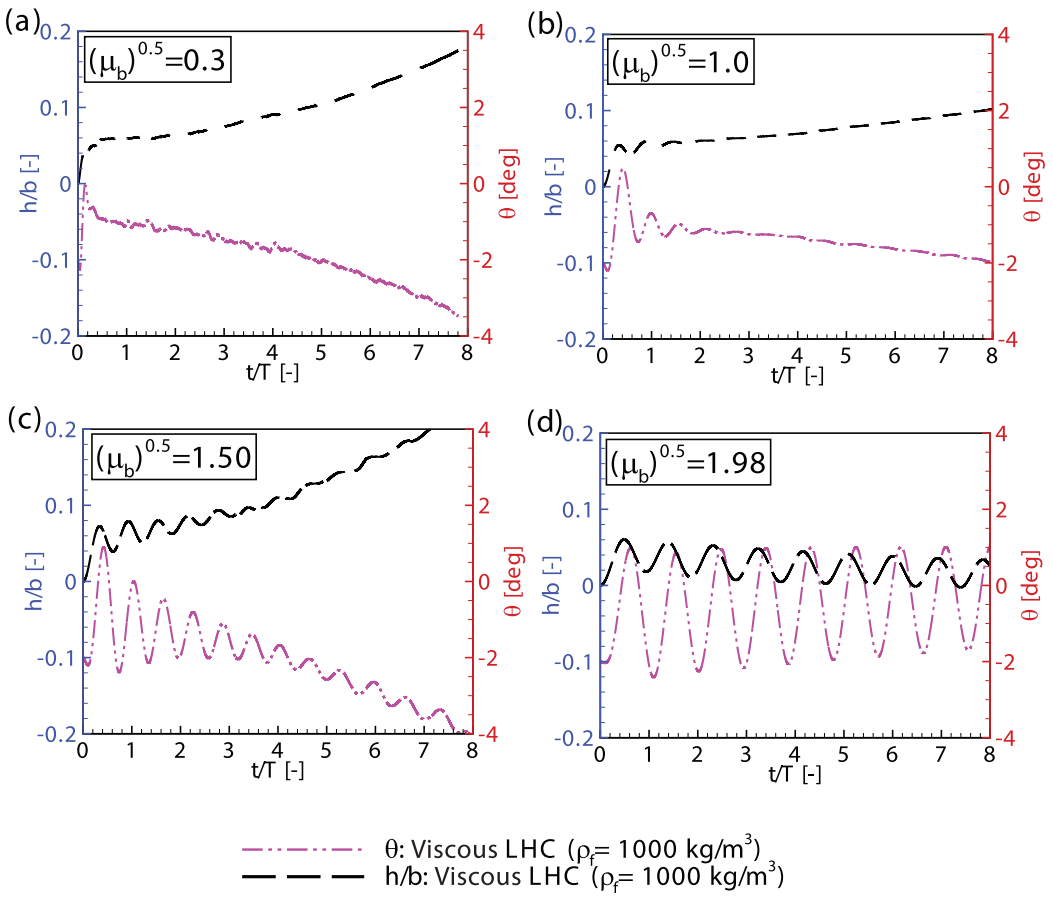


FIG. 9. Time histories of bending and twisting deformation of viscous LHC at (a) $\sqrt{\mu_b} = 0.3$ (: static divergence) (b) $\sqrt{\mu_b} = 1.0$ (: dynamic divergence) (c) $\sqrt{\mu_b} = 1.5$ (: dynamic divergence), (d) $\sqrt{\mu_b}$=1.98 (: flutter). $T = 2\pi/\omega_{\theta-fluid}$ is the natural twisting period of the foil in fluid, which varies with $\sqrt{\mu_b}$. Note that the $\sqrt{\mu_b}$ is varied by fixing ρ_f corresponding to the water but varying ρ_s.

E. FSI responses

This section explains the detailed FSI response of the flexible foil under stable, critical, and unstable conditions at the test points shown in Fig. 10.

The predicted time-histories of the bending and twisting deformations at the test points shown in Fig. 10 and obtained using the time-domain [TD] inviscid FC and viscous LHC method formulation are shown in Figs. 11 and 12; the flutter velocity, U_f, shown in the labels therein corresponds to that computed by the viscous LHC method. As can be seen from Fig. 7, the predicted flutter velocities are somewhat different for the viscous LHC computations and the inviscid FC computations due to the viscous and nonlinear FSI effects. When the inflow velocity is much less than the flutter velocity (as in Figs. 11(a), 11(b), 12(a), and 12(b)), the foil is predicted to be stable by both the inviscid and viscous simulations, where the amplitudes of the bending and twisting vibrations decrease with time and eventually approach zero because of a net positive total damping.

When the inflow velocity is equal to the flutter velocity, the amplitudes of the bending and twisting vibrations remain the same in time due to net zero total damping; this is close to the cases shown in Figs. 11(d), 12(c), and 12(d) for the viscous LHC simulations. When the inflow velocity is much greater than the flutter velocity, as shown in Figs. 11(c), 11(e), 11(f), 12(e), and 12(f), the system becomes unstable and gets into dynamic divergence; the amplitudes of the bending and twisting vibrations increase with time because the total damping becomes negative. In addition, as shown in Fig. 10, the points shown in Figs. 11(e), 11(f), 12(e), and 12(f) are not only above the

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TABLE II. The influence of varying ρ_s vs. ρ_f (to change the solid-to-fluid added mass ratio) on the reduced static/dynamic divergence and flutter velocities. The results are predicted by the viscous LHC model.

		Varying ρ_s ($\rho_f = 1000$ kg/m^3)			Varying ρ_f ($\rho_s = 97.13$ kg/m^3)		
Status	$\sqrt{\mu_b}$	U^*	$\frac{\omega}{\omega_{\theta-fluid}}$	k	U^*	$\frac{\omega}{\omega_{\theta-fluid}}$	k
Static divergence	0.3	0.16	0.0	0.0	...	...	...
	0.67	0.37	0.0	0.0	...	...	...
Dynamic divergence	1.0	0.55	$1.78 \rightarrow 0.0$	$1.25 \rightarrow 0.0$	0.54	$1.37 \rightarrow 0.0$	$1.21 \rightarrow 0.0$
	1.5	0.82	$1.16 \rightarrow 0.0$	$0.81 \rightarrow 0.0$	0.82	$1.22 \rightarrow 0.0$	$0.81 \rightarrow 0.0$
	1.65	0.90	$1.11 \rightarrow 0.0$	$0.73 \rightarrow 0.0$	0.88	$1.19 \rightarrow 0.0$	$0.77 \rightarrow 0.0$
Flutter	1.98	1.00	1.13	0.70	1.02	1.09	0.66
	2.79	...	...	...	1.29	0.93	0.52
	4.18	...	...	...	1.85	0.89	0.36

flutter boundary, but also the static divergence boundary and hence are labeled as dynamic divergence since obvious fluctuations could be observed, the mean deformation increases with time, and the oscillation frequency goes toward zero as deformation increases.

As shown in Fig. 7, for $\sqrt{\mu_b} \leq 1$, the foil will encounter static divergence before flutter. Divergence is possible when the E.A. of the hydrofoil is behind the C.P. (relative to the foil LE). As evident via Eq. (23), the critical divergence velocity decreases with increasing ρ_f and e. However, even for the case of $e = 0$, where the E.A. coincides with the A.C., divergence may still occur, as shown in Fig. 7(b), because viscous effects and flow separation cause the C.P. to move from the theoretical A.C. Also note that for $\sqrt{\mu_b} < 3$, where viscous forces are relatively more important because of the higher fluid density, the results shown in Figs. 11 and 12 suggest that viscous effects may not delay flutter. In this regime, as shown in Figs. 11, 12(a), 12(c), and 12(e), the magnitudes

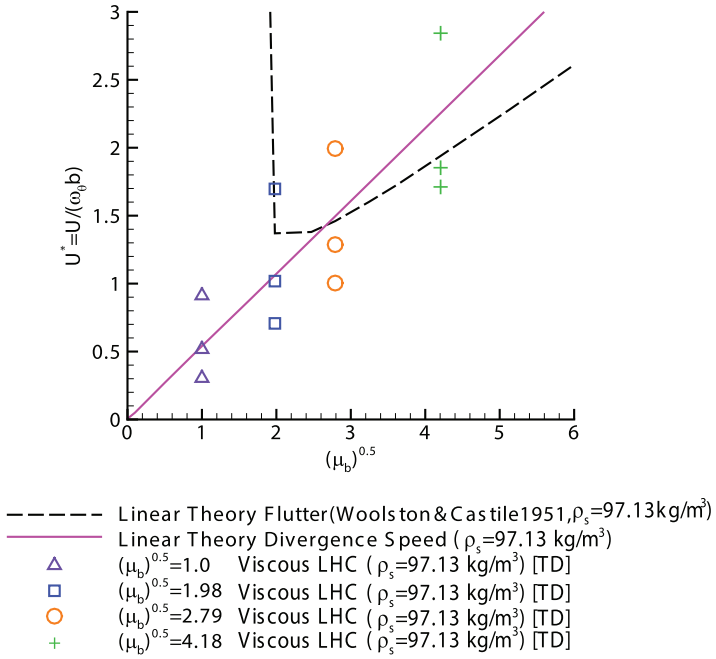


FIG. 10. The test points used to demonstrate the detailed FSI response of flexible foils in light and dense flow conditions. Also shown are the linear theory flutter (via frequency-domain solution method) and static divergence velocity boundaries. The problem set-up is similar to the experiments by Woolston and Castile.[20]

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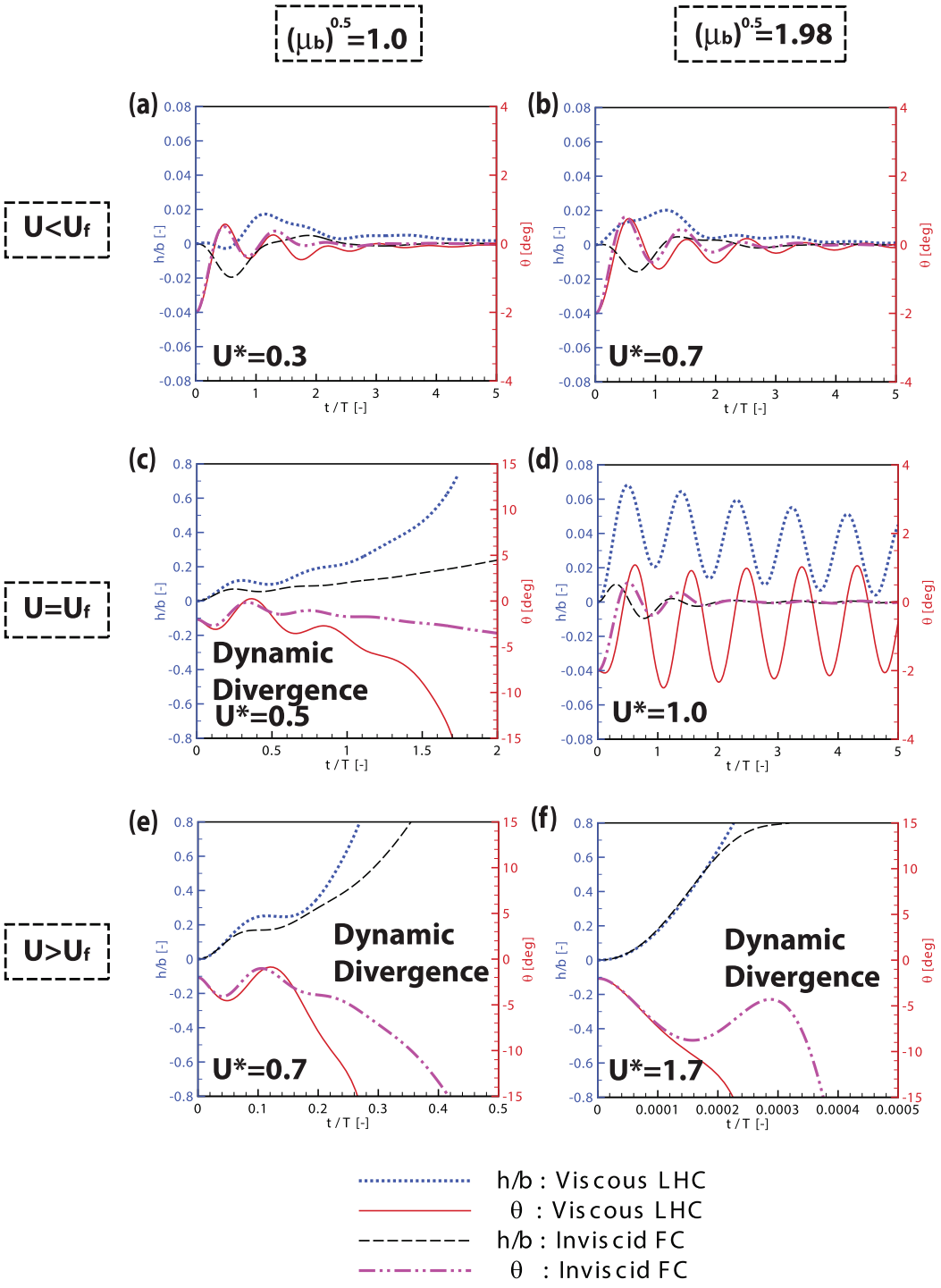


FIG. 11. (a–f) Time-histories of bending and twisting deformations at $\sqrt{\mu_b} = 1.0$ and $\sqrt{\mu_b} = 1.98$. $T = 2\pi/\omega_{\theta\,-\,fluid}$ is the natural twisting period of the foil in fluid, $\sqrt{\mu_b}$ is varied by fixing ρ_s corresponding to the Balsa wood foil used by Woolston and Castile,[20] but varying ρ_f.

of the bending and twisting deformations obtained using the viscous LHC simulations are greater than the inviscid FC simulations, due to the nonlinear viscous FSI effect caused by the higher fluid density. Consequently, the predicted flutter velocity using the viscous LHC simulations is lower than the inviscid FC simulations, as will be shown later in Fig. 17. However, for $\sqrt{\mu_b} > 3$, the results

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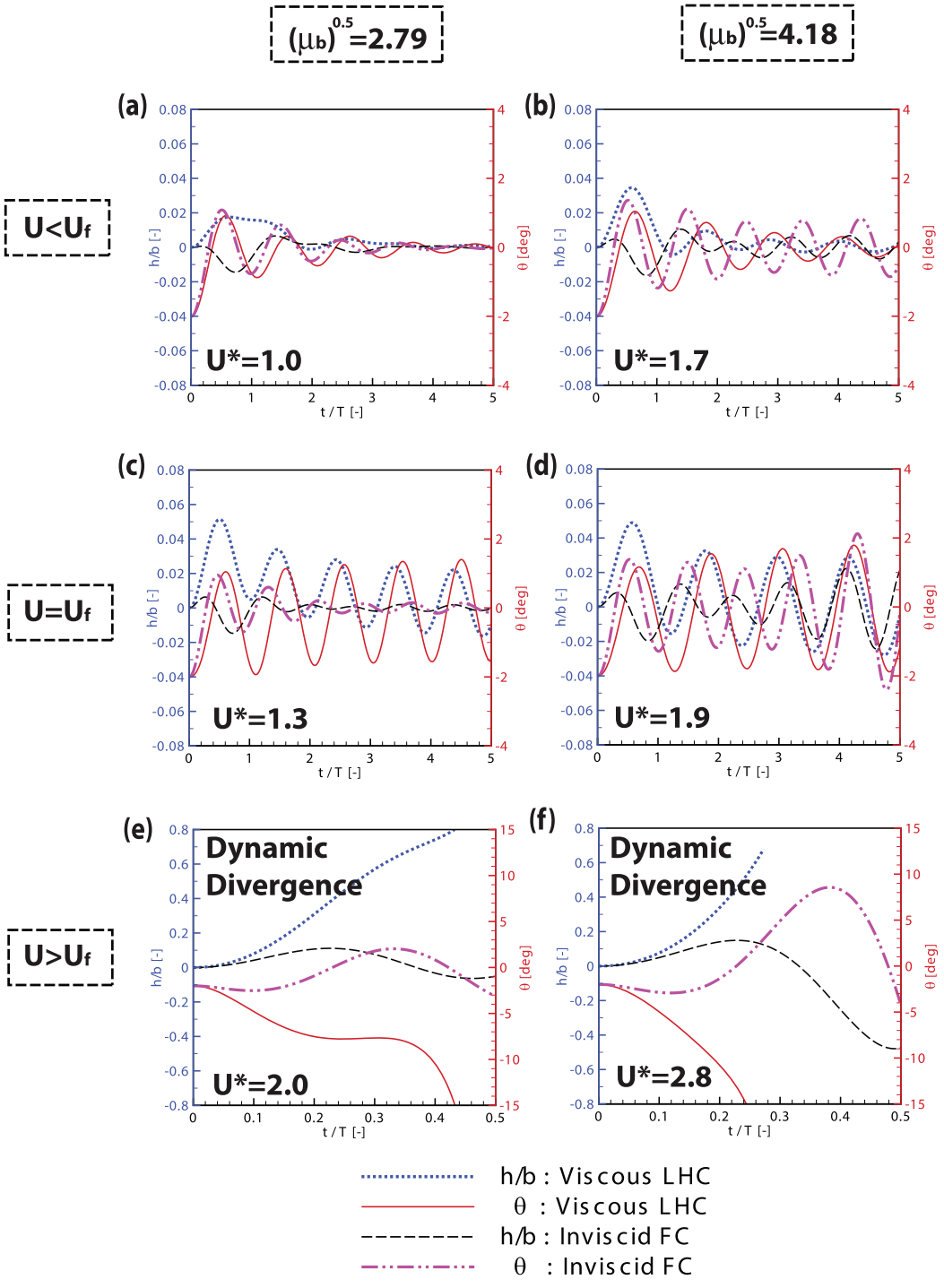


FIG. 12. (a–f) Time-histories of bending and twisting deformations at $\sqrt{\mu_b} = 2.79$ and $\sqrt{\mu_b} = 4.18$. $T = 2\pi/\omega_{\theta-fluid}$ is the natural twisting period of the foil in fluid, $\sqrt{\mu_b}$ is varied by fixing ρ_s corresponding to the Balsa wood foil used by Woolston and Castile,[20] but varying ρ_f.

in Fig. 7 show that viscous effects help to delay flutter. For instance, the inviscid FC simulations predict a lower flutter velocity than the viscous LHC simulations, which is evident by the increasing deformation amplitudes with time predicted by the inviscid FC method in Fig. 12(d), i.e., $U > U_f$, while the viscous LHC method shows approximately constant displacement amplitude in time.

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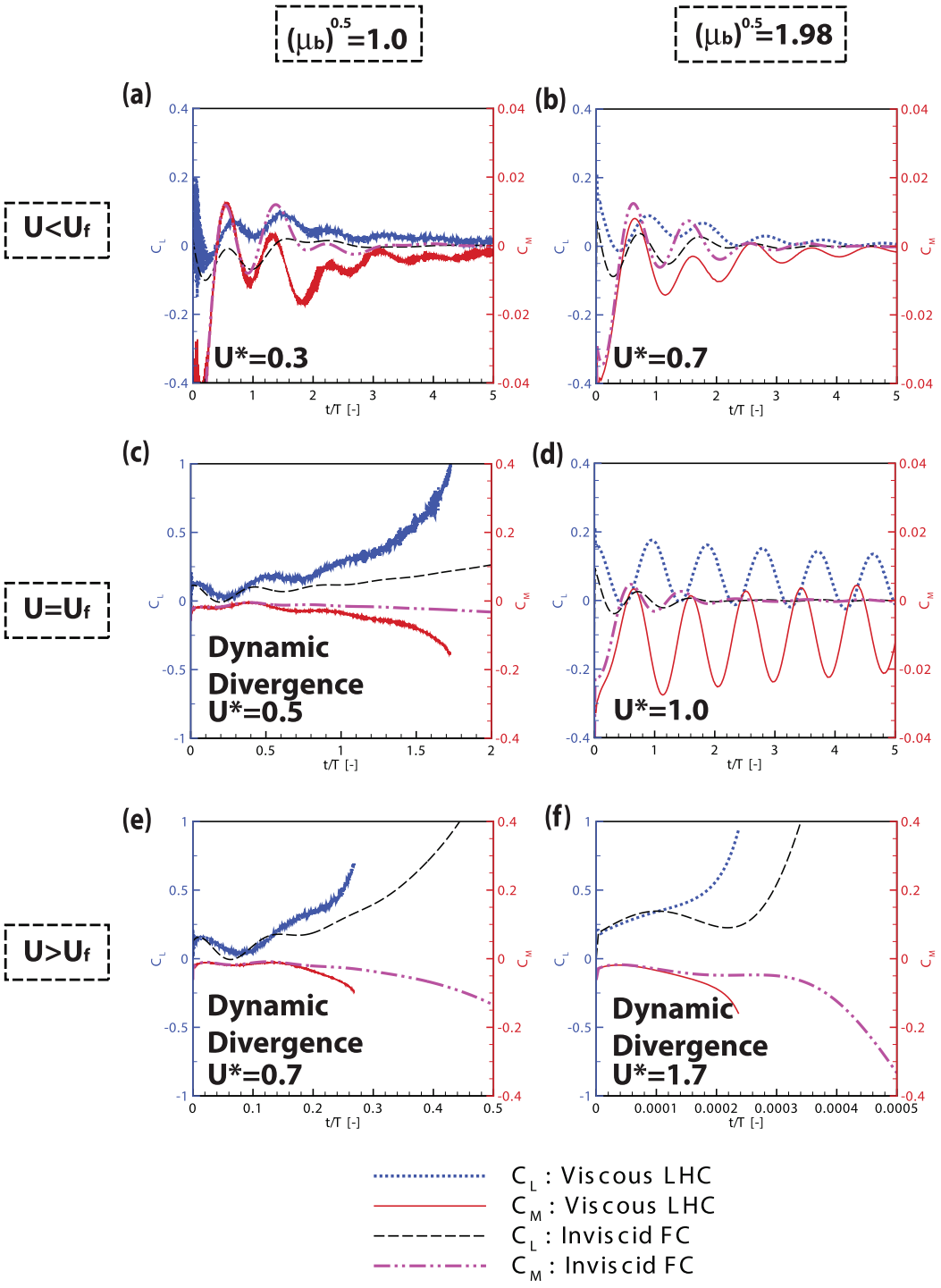


FIG. 13. (a–f) Time-histories of lift and moment coefficients at $\sqrt{\mu_b} = 1.0$ and $\sqrt{\mu_b} = 1.98$. $T = 2\pi/\omega_{\theta\,-\,fluid}$ is the natural twisting period of the foil in fluid, $\sqrt{\mu_b}$ is varied by fixing ρ_s corresponding to the Balsa wood foil used by Woolston and Castile,[20] but varying ρ_f. Note that for all the cases except for (a), the results are obtained using $\Delta t = 4 \times 10^{-4}$ s. For case (a) $\Delta t = 8 \times 10^{-4}$ s is used in order to minimize numerical noises due to the relatively large period (T).

The time-histories of the lift coefficient ($C_L = L/(\frac{1}{2}\rho_f U^2 c)$) and moment coefficient ($C_M = M/(\frac{1}{2}\rho_f U^2 c^2)$) for various solid-to-fluid added mass ratios for the system under stable, critical, and unstable conditions are shown in Figs. 13 and 14. Note that the lift coefficients follow

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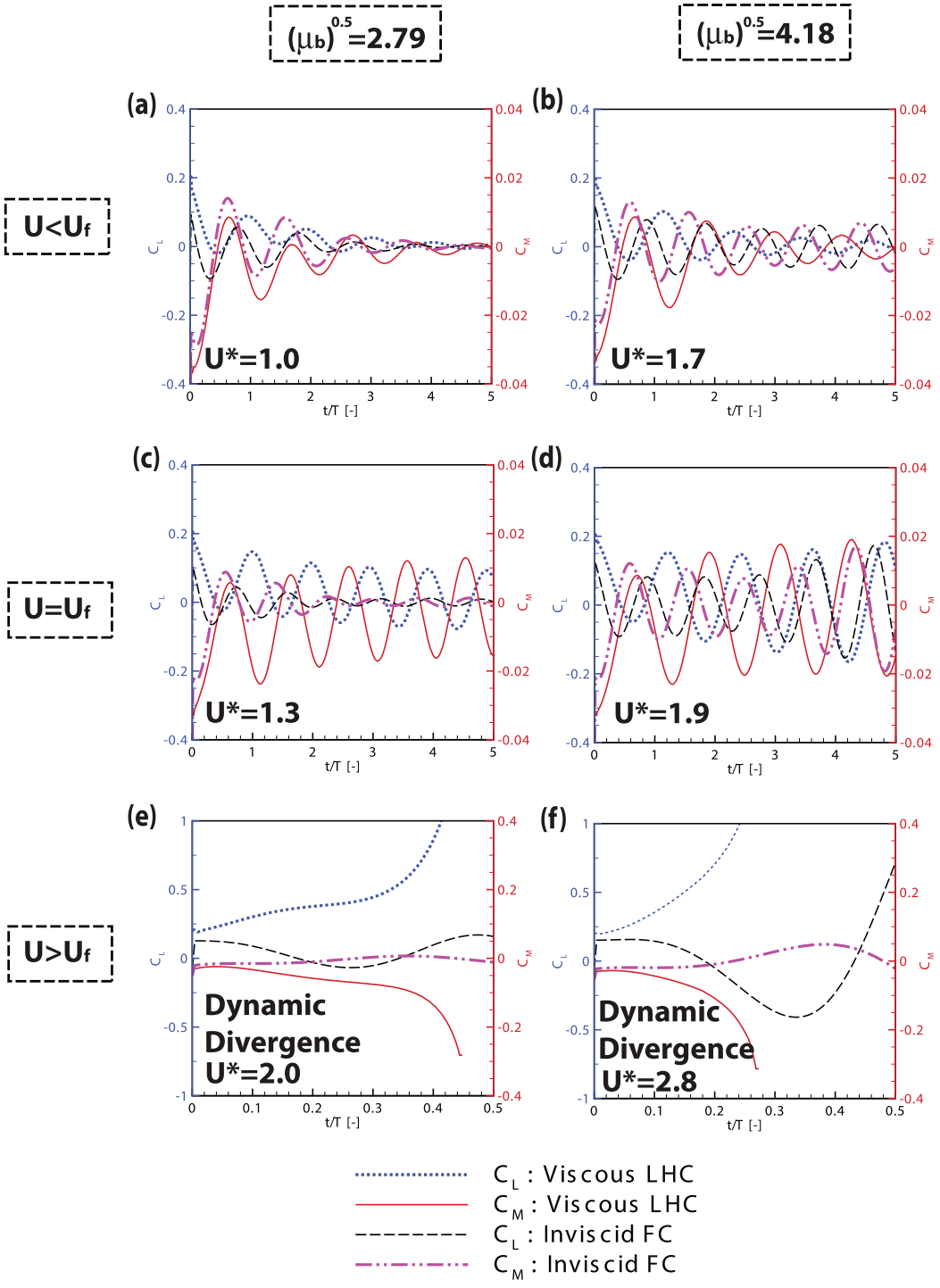


FIG. 14. (a–f) Time-histories of lift and moment coefficients at $\sqrt{\mu_b} = 2.79$ and $\sqrt{\mu_b} = 4.18$. $T = 2\pi/\omega_{\theta-fluid}$ is the natural twisting period of the foil in fluid, $\sqrt{\mu_b}$ is varied by fixing ρ_s corresponding to the Balsa wood foil used by Woolston and Castile,[20] but varying ρ_f.

the same trend with the bending deformations; similarly, the moment coefficients (defined positive counterclockwise about the E.A.) follow the same trend with the twisting deformations. For $\sqrt{\mu_b} \leq 1$, the C_L and C_M predictions obtained using the viscous LHC simulations have high frequency noise introduced by the relatively small time-step (compared to the large natural period). The

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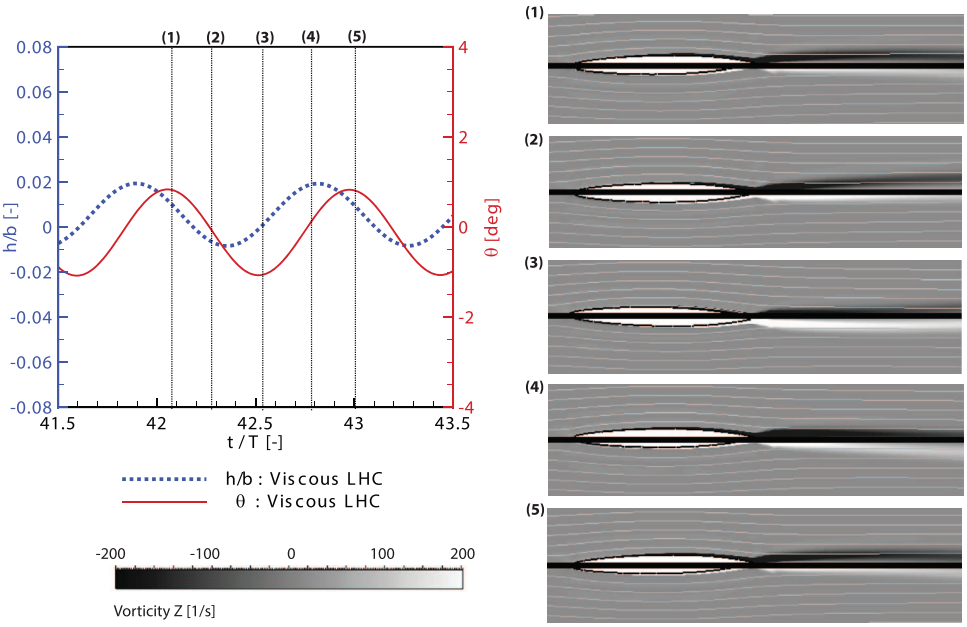


FIG. 15. (Left) Time-histories of the bending and twisting deformation and (right) the predicted vorticity contour with sample streamlines, for the Balsa wood foil used by Woolston and Castile[20] at its flutter velocity ($U^* = 1.0$) with $\sqrt{\mu_b} = 1.98$ ($\rho_f = 2.21$ kg/m^3 and $\rho_s = 97.13$ kg/m^3) and Re $= 1.94 \times 10^6$. $T = 2\pi/\omega_{\theta\ -\ fluid}$ is the natural twisting period of the foil in fluid. The heavy black horizontal lines on the vorticity contour plot indicate the initial vertical position of the foil trailing edge.

large natural period is a result of the higher added mass with higher ρ_f. To minimize this artificial noise due to numerical errors, the simulation shown in Fig. 13(a) for the lift and moment coefficients with $U^* = 0.3$ and $\sqrt{\mu_b} = 1.0$ was obtained using $\Delta t = 8 \times 10^{-4}$ s.

Figure 15 illustrates the predicted time-history of the bending and twisting deformations along with selected snapshots of the predicted streamlines and vorticity contours obtained from the viscous LHC simulations for a hydrofoil at its critical flutter velocity of $U^* = 1.0$ for $\sqrt{\mu_b} = 1.98$, for the Balsa wood foil used by Woolston and Castile.[20] The heavy black horizontal lines on the vorticity contour plot indicate the undeformed vertical position of the hydrofoil trailing edge (T.E.). The deformations are relatively small because of the relative high stiffness of the model-scale foil. Nevertheless, the foil is observed to undergo near constant amplitude oscillations in plunge and pitch at the flutter velocity.

Similarly, Fig. 16 shows the predicted deformation time-histories and corresponding snapshots of the streamline and vorticity contours for a hydrofoil at the divergence velocity for $\sqrt{\mu_b} = 1.98$ with $U^* = 1.7$. For speeds greater than the divergence velocity ($U^* \geq 1.7$), the amplitudes of both the bending and twisting motion increase monotonically with time.

Note that the vorticity contours in Figs. 15 and 16 show consistent thin trailing wake sheets; this is because $k < 1$, which is consistent with the wake patterns reported in Munch, Ausoni, and Braun.[14]

F. Total damping coefficient and vibration frequency

In this study, the flutter boundaries are determined as the critical speeds when the total bending (or twisting) damping coefficients, ζ_t, are projected to be zero. To illustrate this process, the predicted total bending and twisting damping coefficients, $\zeta_{t\text{-bending}}$ and $\zeta_{t\text{-twisting}}$, as a function of the (dimensional) flow speed are shown in Fig. 17 along with the measured flutter velocity values reported by Woolston and Castile.[20]

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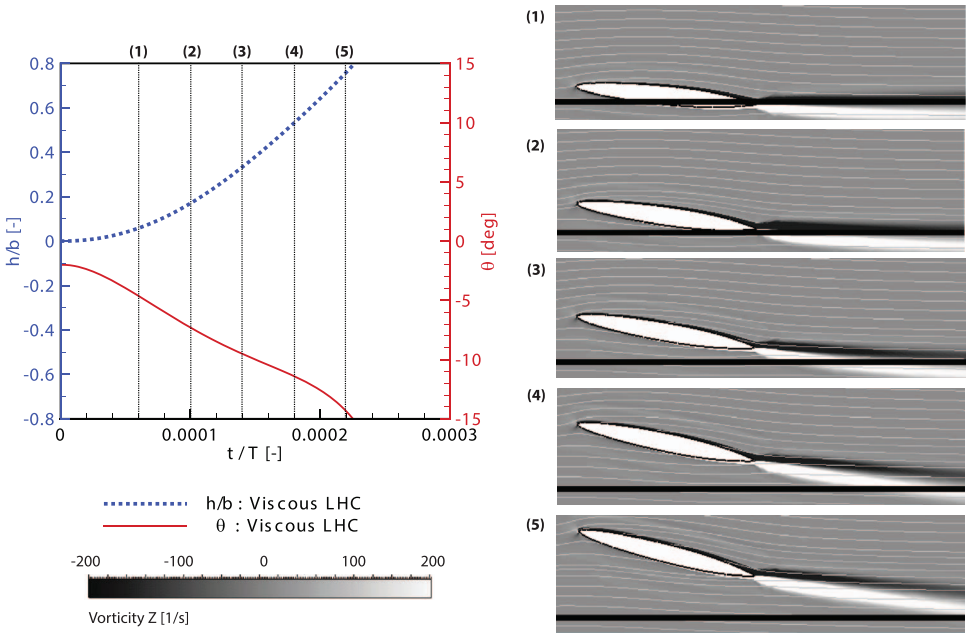


FIG. 16. (Left) Time-histories of the bending and twisting deformation and (right) the predicted vorticity contour with sample streamlines, for the Balsa wood foil used by Woolston and Castile[20] at its divergence velocity ($U^* = 1.7$) with $\sqrt{\mu_b} = 1.98$ ($\rho_f = 2.21$ kg/m^3 and $\rho_s = 97.13$ kg/m^3) and Re $= 3.24 \times 10^6$. $T = 2\pi/\omega_{\theta\ -\ fluid}$ is the natural twisting period of the foil in fluid. The heavy black horizontal lines on the vorticity contour plot indicate the initial vertical position of the foil trailing edge.

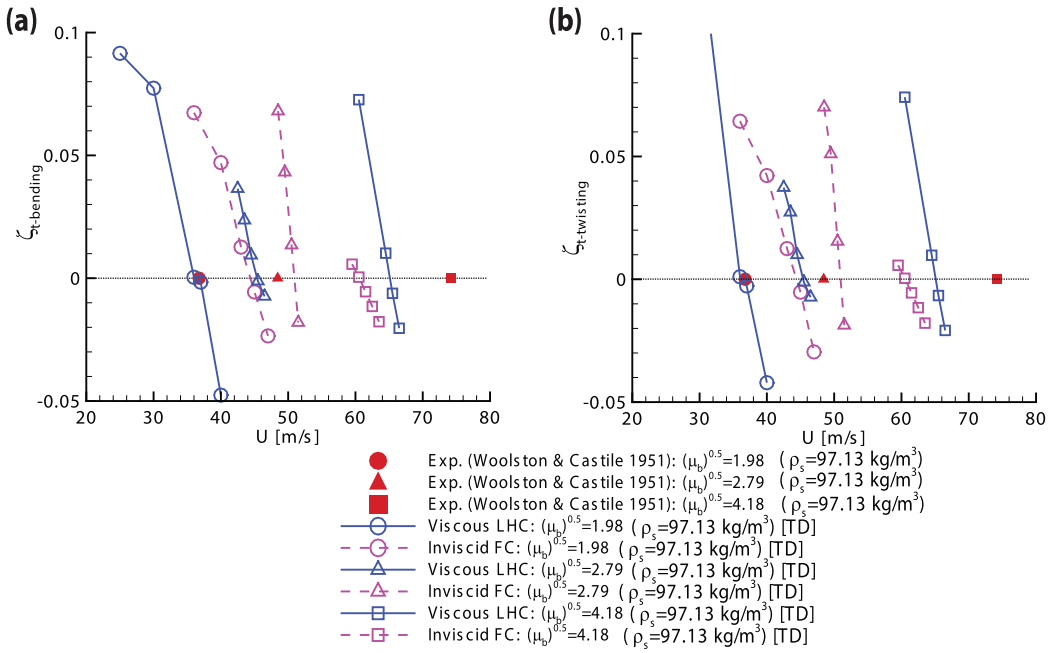


FIG. 17. (a) Total bending damping ratio ($\zeta_{\text{t-bending}}$), and (b) total twisting damping ratio ($\zeta_{\text{t-twisting}}$) as function of the flow velocity (U) at different solid-to-fluid solid-to-fluid added mass ratios $\sqrt{\mu_b}$. In this plot $\sqrt{\mu_b}$ is varied by fixing ρ_s corresponding to the Balsa wood foil used by Woolston and Castile,[20] but varying ρ_f. The filled symbols correspond to the experimental measurements, the dashed lines with open symbols correspond to inviscid FC simulations, and the solid lines with open symbols correspond to viscous LHC simulations.

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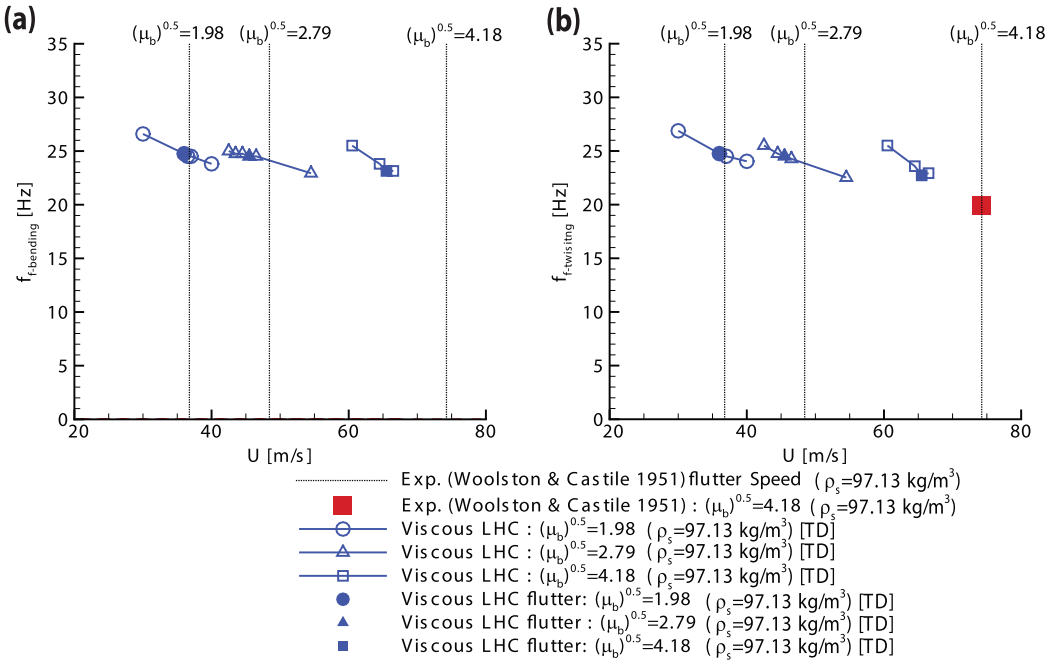


FIG. 18. (a) Bending frequency ($f_{\text{f-bending}}$) and (b) twisting frequency ($f_{\text{f-twisting}}$) at different solid-to-fluid added mass ratios $\sqrt{\mu_b}$ at the various simulation points shown in Fig. 10. $\sqrt{\mu_b}$ is varied by fixing ρ_s corresponding to the Balsa wood foil used by Woolston and Castile[20] but varying ρ_f. Also shown are the measured flutter velocity (as shown by the vertical lines) and frequency (large filled square) from Woolston and Castile.[20] The experimental frequencies for different $\sqrt{\mu_b}$ are also shown in Fig. 8. The flutter velocities are noted by the filled symbols.

As noted earlier, when $\zeta_t > 0$, the deformations will decrease with time, so the system is stable. When $\zeta_t < 0$, the deformations will increase with time, so the system is unstable. The flutter velocities reported in Fig. 7 correspond to the projected critical speed in Fig. 17 at which either $\zeta_{t\text{-bending}} = 0$ or $\zeta_{t\text{-twisting}} = 0$, whichever is earlier, although the two values are very close.

As shown in Fig. 17, the flutter velocity predicted by the viscous LHC method is higher than that the predicted velocities by the inviscid FC method for $\sqrt{\mu_b} > 3$, but the opposite is true for $\sqrt{\mu_b} < 3$. Moreover, for all the cases shown, the flutter velocities predicted by the viscous LHC method are closer to the experimentally measured values than those predicted by the inviscid FC method. It should be noted that the current simulations assume the structural damping coefficients, ζ_h and ζ_θ in Eq. (1), to be zero (for simplicity), which contributes to the slight under prediction of the flutter velocities predicted by the viscous LHC method compared to the experimentally measured values.

Figure 18 shows the predicted system bending and twisting vibration frequencies (shown with open symbols) as a function of the flow speed for three different relative mass ratios. The predictions shown there are obtained using the viscous LHC method. It should be noted that results for the $\sqrt{\mu_b} = 1$ case are not shown in Fig. 18, because in that case divergence occurred prior to flutter. Note that although the measured flutter velocities were reported for multiple cases, the measured flutter frequency was only reported for the $\sqrt{\mu_b} = 4.18$ case in Woolston and Castile.[20] Additional comparisons of the predicted and measured non-dimensional flutter frequencies are shown in Fig. 8, which have different $\sqrt{\mu_b}$ values than the ones shown here. The results show that the viscous LHC simulations are in good agreement with the measured values.

IV. CONCLUSIONS

In this paper, the FSI response and stability of a flexible foil in dense, turbulent, and incompressible flow were investigated using an URANS fluid solver coupled with a 2-DOF solid solver

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through an efficient and stable LHC method introduced by Young, Chae, and Akcabay.[30] The results were compared with the predictions of a FC linear potential theory formulation,[10] as well as with several published experimental data given by Woolston and Castile[20] and Besch and Liu.[27] The foil geometry was that of NACA 16-010, which is the same as that used in the experimental study by Woolston and Castile.[20] The operating conditions ranged from $\mathrm{Re} = 3.05 \times 10^5 - 4.27 \times 10^6$ and $\sqrt{\mu_b} = 0.3$–4.18. The LHC simulations converged without any numerical stability issues for all the cases.

The results demonstrated the influence of the fluid added mass, hydrodynamic damping, and nonlinear FSI response on the foil stability. In general, as the fluid density increased for a given solid density or as the solid density decreased for a given fluid density (i.e., decreasing solid-to-fluid added mass ratio, $\sqrt{\mu_b}$), both the critical reduced flutter and reduced divergence velocities were observed to decrease. It should be noted here that the reduced velocities are non-dimensionalized by $\omega_\theta b$, where ω_θ is the natural twisting frequency of the foil in vacuum and b is the semi-chord of the foil. For a given foil with solid density (ρ_s), both dimensional flutter and divergence velocities decrease with increasing fluid density (ρ_f), i.e., the same foil will encounter flutter/divergence earlier in water compared to air. However, the trend is not as obvious for cases with fixed ρ_f but varying ρ_s, because ω_θ (which is used to define the reduced velocity) varies with $1/\sqrt{\rho_s}$.

The results also showed that the frequency-domain linear potential theory solution predicts higher flutter velocities than measured, i.e., unconservative, for cases with low solid-to-fluid added mass ratios ($\sqrt{\mu_b} < 3$). This is because in the low solid-to-fluid added mass ratio regime, the fluid forces are comparable to the solid forces, which lead to strong, nonlinear, and viscous interactions. The results also showed that for $\sqrt{\mu_b} < 1$, static divergence might occur prior to flutter for light hydrofoils even if the E.A. is at the aerodynamic center (theoretical center of pressure) based on the linear potential theory for thin symmetric foils. This is because the C.P. varies with angle of attack and flow conditions due to viscous and flow separation effects, which leads to a non-zero moment arm between the E.A. and the C.P., leading to the possibility of divergence, which could only be captured by viscous FSI models. In addition, dynamic divergence, which may occur before flutter at the low solid-to-fluid added mass ratio regime ($1 \leq \sqrt{\mu_b} < 2$), can only be captured by viscous nonlinear FSI models because of the greater relative contribution of nonlinear viscous FSI effects. For $\sqrt{\mu_b} > 2$, however, flutter tends to occur before divergence, and the difference between inviscid and viscous models become relatively small, as both of them agree well with experimental measurements.

In general, the predicted flutter velocity, flutter frequency, and divergence velocity obtained using the viscous LHC simulations compared well with the measured values from previously published experimental data. The predicted flutter velocity obtained using the viscous LHC method was slightly lower than the measured values; this might be because structural damping was ignored in the current computations. The results showed that the flutter velocity predicted using the inviscid FC solver was lower than the viscous LHC solver for $\sqrt{\mu_b} > 3$, but the opposite was true for $\sqrt{\mu_b} < 3$. The reason for the general good agreement on the divergence velocity using both the linear potential theory, inviscid FC and viscous LHC simulations ("general," because a case was just discussed above for where the linear potential theory predicted an infinite divergence velocity, but the actual measured divergence velocity was finite, and was captured only using the viscous LHC method), as well as experimental measurements, in the low solid-to-fluid added mass ratio regime was because the nonlinear viscous FSI were mostly associated with the fluid inertial and damping terms, but static divergence is a static failure that concerns the stiffness disturbing force terms only. Even though the inviscid FC method with a time-domain solution technique provides a reasonable approximation of the flutter and static divergence velocities at a much lower computational cost compared to the viscous LHC method, viscous LHC simulations in time-domain are recommended to predict the dynamic response and stability boundary because of important nonlinear viscous FSI effects. The need to use viscous FSI solvers is particularly necessary for the region governed by dynamic divergence ($1 \leq \sqrt{\mu_b} < 2$), which is a nonlinear phenomenon that cannot be predicted by the linear methods.

For future work, the authors are working on a follow-up study that: (1) derives and quantifies the influence of key fluid, material, and geometric parameters (including the location of the elastic

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axis) that govern the hydroelastic response and stability of flexible lifting surfaces, and (2) develops useful parametric maps with common materials and dimensions to assist the design of flexible lifting surfaces, especially for marine applications. The authors will also extend the new LHC method to simulate the transient FSI response and stability of 3D hydrofoils and propulsors. Additionally, the authors will investigate the effects of fluid cavitation, free surface, material anisotropy, and scaling laws on the dynamic viscous FSI response and stability of flexible hydrofoils.

ACKNOWLEDGMENTS

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